TanGCE: Manifold-Aware Concept Erasure

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Abstract

Concept erasure is useful only when it removes a target attribute without destroying the other information encoded in the representation. This is hard in the nonlinear setting: information erased for one probe may remain recoverable by a fresh nonlinear probe. Direct gradient-based edits introduce a second failure mode: they may remove target signal by pushing representations *off the manifold* of natural representations, damaging the many other concepts encoded along that manifold — most of which are unlabeled and cannot be explicitly protected as controls. Our central argument is that nonlinear erasure becomes surgical precisely when each edit is constrained to stay on the natural-representation manifold: an on-manifold step removes target signal in directions the representation itself supports, so collateral damage to unknown, unlabeled control concepts is bounded by construction. We propose the TanGCE family, a manifold-aware approach to nonlinear concept erasure that operationalises this argument. The core method, TanGCE, is an iterative manifold-constrained step: for each representation, nearest-neighbor differences estimate the local tangent space of the manifold, and a fresh nonlinear scorer's gradient is projected onto that tangent, scaled by the local manifold extent, and capped by a per-sample trust region. Two preprocessing-augmented variants compose this core with closed-form linear projections: TanGCE^{+} prepends LEACE (mean alignment between target classes), and TanGCE^{++} additionally prepends a rank-2 covariance projection that exhausts the second-moment target signal. We evaluate the family on 13 language models, three NLP concepts, and 40 CelebA-CLIP attributes under two control regimes, using one fixed set of hyperparameters with no per-setting tuning. Compared at the same allowed drop in control-concept accuracy, TanGCE^{++} removes more target signal than the strongest published baseline. On the NLP settings, it drives a fresh nonlinear probe to the majority-class chance level in 26–33 settings, 13–20 more than the baseline, while causing smaller control degradation. The advantage is largest when target and control attributes are entangled: on CelebA-CLIP, under highly correlated controls, TanGCE^{++} is the only method that stays within the allowed control-accuracy drop for every target attribute — the empirical signature of an erasure step that respects the manifold the controls also live on.

1 Introduction

Neural representations encode many attributes of their inputs: semantic content, style, demographic information, task labels, and other latent factors. In many applications, we may want to remove one such attribute, for example a protected or spurious concept (e.g., gender and toxicity), while preserving everything else the representation contains [De-Arteaga et al., 2019, Avitan et al., 2025]. This is the central tension in *concept erasure*: removing the target information while unnecessarily damaging other control concepts, known as *surgicality* Singh et al. [2024].

The surgicality requirement makes erasure especially difficult. A target concept is rarely encoded in isolation. It may be entangled with many other factors, most of which are unlabeled and therefore cannot be explicitly protected. Moreover, erasure must address *recoverability*: even when a chosen classifier is driven to chance, a freshly trained classifier on the edited representations may still recover the protected attribute [Elazar and Goldberg, 2018]. Linear erasure methods provide simple and

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effective closed-form interventions [Ravfogel et al., 2020, Belrose et al., 2023], but they only remove linearly accessible information and do not generally block nonlinear probes [Ravfogel et al., 2022]. Nonlinear erasure, however, introduces a different danger: if we directly follow the gradient of a nonlinear concept scorer in the ambient representation space, the edit may remove target signal by moving the representation to a region that natural inputs never occupy.

To address this, we propose the *Manifold Representation Hypothesis* (MRH; §4): natural representations of a trained model concentrate on a low-dimensional manifold $\mathcal{M} \subset \mathbb{R}^d$, and an erasure step that moves each representation *along* $\mathcal{M}$ is structurally surgical, while a step with off-manifold components damages the unlabeled control concepts that ride along the same manifold. MRH makes a directly testable prediction: if the on-manifold constraint is what makes erasure surgical, then bolting it on top of *any* prior erasure pipeline should reduce residual leakage without further damaging the controls. We confirm exactly this in §7.2, where stacking TanGCE on top of five prior erasers (LEACE, LEACE + CovMatch, INLP, IGBP, Obliviator) lowers nonlinear-probe leakage on every panel while preserving control accuracy.

We introduce the TanGCE family, a manifold-aware approach to nonlinear concept erasure that operationalises this claim. The core method, TanGCE, is an iterative manifold-constrained loop: for each representation, we approximate the manifold locally by its first-order tangent space at that point, estimated via SVD on nearest-neighbor secants [Zhang and Zha, 2004]. We project the concept-scorer gradient onto this tangent space, restricting the edit to directions the manifold actually extends along. We then rescale each tangent coordinate by the corresponding local singular value, a local-σ preconditioner that ties the step magnitude in each direction to how far the manifold extends along that axis. Two preprocessing-augmented variants compose this core with closed-form linear stages: TanGCE$^+$ prepends LEACE [Belrose et al., 2023], exhausting the first-moment mean difference; TanGCE^{++} further prepends a rank-2 projection that removes the leading second-moment covariance asymmetry. The combined effect is that every nonlinear edit is an on-manifold step: target signal is removed, and the unlabeled control concepts that ride along the same manifold are protected by construction.

We test the TanGCE family on text and vision tasks: 13 language models spanning 0.5B–27B parameters on three target concepts (sycophancy, gender bias, safety), and CLIP image embeddings on 40 face-attribute concepts under two control regimes. We operationalize surgicality on every setting by fixing a *control-damage budget*, the maximum allowed drop in held-out control-concept probe accuracy, and reporting how much target-concept signal each method removes within it (measured by a fresh nonlinear probe). TanGCE^{++} (the full pipeline: LEACE + rank-2 CovMatch + manifold loop) is the headline configuration on both the NLP grid and CelebA, beating the strongest published baseline at every budget under one fixed set of hyperparameters with no per-setting tuning.

Our contributions are:

1. We name and empirically test the *Manifold Representation Hypothesis* (MRH): that on-manifold edits are surgical by construction. Beyond our own pipeline, we show that adding the manifold-constrained step on top of five unrelated prior erasers (LEACE, LEACE + CovMatch, INLP, IGBP, Obliviator) reduces residual leakage while preserving control accuracy (§7.2) — direct evidence that the manifold constraint, not any one preprocessor, is doing the work.
2. We introduce the TanGCE family: a manifold-aware iterative erasure step (TanGCE), and two preprocessing-augmented variants (TanGCE$^+$ = LEACE + TanGCE; TanGCE^{++} = LEACE + rank-2 CovMatch + TanGCE) that ablate the contribution of each closed-form preprocessing stage.
3. We propose a local-σ preconditioner that scales tangent-coordinate steps by the local singular spectrum, making the trust region sensitive to the manifold's anisotropic geometry.

2 Related Work

Concept erasure methods differ mainly in the function class against which they erase and in the geometry they assume. Linear methods such as INLP [Ravfogel et al., 2020] and LEACE [Belrose et al., 2023] target linearly decodable concept information. INLP removes successively exposed linear directions by iterative nullspace projection, while LEACE gives the strongest closed-form linear

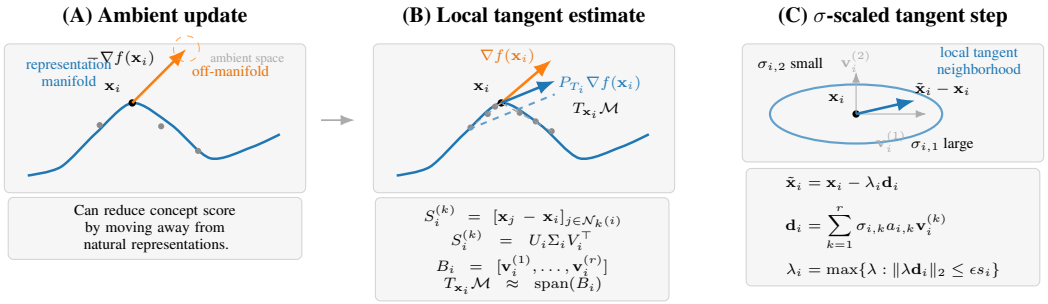


Figure 1: **Intuition for TanGCE. (A)** An ambient nonlinear erasure step can reduce concept score by moving a representation away from the support of natural inputs. **(B)** TanGCE estimates a local tangent space $T_{\mathbf{x}_i}\mathcal{M}$ from k nearest-neighbor secants and projects the scorer gradient onto it. **(C)** The projected update is scaled by the local singular spectrum, so motion is larger along well-supported tangent directions and smaller along thin, fragile ones, while remaining inside a local trust region.

baseline in our comparisons. These methods are essential reference points, but they do not directly address the case where a fresh nonlinear probe can still recover the concept after linear erasure.

That recoverability concern is central to our framing. Elazar and Goldberg [Elazar and Goldberg, 2018] showed that apparent success against one learned remover or classifier does not imply that the protected information is gone for a newly trained downstream model. Amnesic probing [Elazar et al., 2021] and related intervention-based analyses adopt the same conservative lesson: the meaningful question is what remains recoverable after the intervention, not only what the original scorer no longer captures.

Two nonlinear erasure methods anchor our comparisons. IGBP [Iskander et al., 2023] is the closest prior baseline in mechanism: it iteratively trains a nonlinear probe and retracts representations toward the current decision boundary. Obliviator [Akbari et al., 2025] is the strongest published baseline in headline numbers: it formulates erasure in RKHS with a Hilbert–Schmidt independence criterion (HSIC) penalty between the representation and the target attribute, and gradually morphs the feature space via random Fourier feature eigenvalue projection (RFF-EVP). Both methods are supervised on the target concept and unsupervised on control concepts (the same regime as our method); they differ from TanGCE in how the update is constructed, not in what supervision they use. We report INLP, IGBP, and Obliviator under the same trajectory-probe protocol as every other method in Tab. 1. Our paper differs in two ways. First, AmbGCE^{++} is introduced only as the direct ambient-space ablation that keeps the iterative nonlinear scorer loop and the same closed-form preprocessing but removes the manifold constraint. Second, the TanGCE family is the paper's main contribution: each variant projects the erasure update onto a locally estimated tangent space of the representation manifold, making the geometric hypothesis itself the main point of comparison.

3 Problem setup and geometric motivation

Let p_{nat} be a distribution over natural inputs $s \in \Sigma^*$, and let $\text{enc}_\ell : \Sigma^* \to \mathbb{R}^d$ be the layer-ℓ encoder of a language model. For a random natural input $S \sim p_{\text{nat}}$, define the layer-ℓ representation random vector

$$\mathbf{X}_\ell = \text{enc}_\ell(S) \in \mathbb{R}^d. \tag{1}$$

Let p_ℓ denote the distribution of $\mathbf{X}_\ell$ over representation space. We define the *natural representation support* as

$$\mathcal{S}_\ell = \text{supp}(p_\ell) = \bigcap_{\substack{A\subseteq\mathbb{R}^d,\ A \text{ closed} \\ \Pr(\mathbf{X}_\ell\in A)=1}} A. \tag{2}$$

Informally, $\mathcal{S}_\ell$ is the region of representation space occupied by hidden states produced by natural inputs.

For each observed sample i, we have a representation $\mathbf{x}_i = \text{enc}_\ell(s_i) \in \mathbb{R}^d$, a binary concept label $y_i \in \{0,1\}$, and an optional control label z_i. The dataset is $\mathcal{D} = \{(\mathbf{x}_i, y_i, z_i)\}_{i=1}^N$, with class subsets $\mathcal{D}_c = \{i : y_i = c\}$. A concept scorer $f_\theta : \mathbb{R}^d \to \mathbb{R}$ is a 2-layer MLP with input gradient $\nabla f_\theta(\mathbf{x}_i)$.

Local manifold hypothesis. Although $\mathbf{X}_\ell$ takes values in the ambient space $\mathbb{R}^d$, we do not expect p_ℓ to occupy this space uniformly. We assume that, around observed natural representations, the support $\mathcal{S}_\ell$ is locally well approximated by a low-dimensional manifold-like structure $\mathcal{M}_\ell \subset \mathbb{R}^d$. The operational consequence is that the neighborhood of a point $\mathbf{x} \in \mathcal{S}_\ell$ can be approximated by a tangent space $T_{\mathbf{x}}\mathcal{M}_\ell$ [Tu, 2011], allowing us to apply local linear operations such as SVD, projection, and anisotropic scaling.

The tangent space is not observed directly. We estimate it from nearby representations by local SVD on nearest-neighbor secants (§5). For an ambient vector $\mathbf{g} \in \mathbb{R}^d$, the tangent decomposition

$$\mathbf{g} = \mathbf{g}^\top + \mathbf{g}^\perp, \qquad \mathbf{g}^\top \in T_{\mathbf{x}}\mathcal{M}_\ell, \qquad \mathbf{g}^\perp \perp T_{\mathbf{x}}\mathcal{M}_\ell, \tag{3}$$

separates the component supported by the local representation geometry from the component orthogonal to it. TanGCE acts on the tangent component only; AmbGCE^{++} retains both. Formal coordinate-chart definitions and the differential $d(\varphi^{-1})_0$ identification with $T_{\mathbf{x}}\mathcal{M}_\ell$ are deferred to App. I.

Recoverability. Concept removal is judged only after training a *fresh* probe on the edited representations [Elazar and Goldberg, 2018]: erasure succeeds if a newly trained nonlinear classifier fails to recover the concept from $\tilde{\mathbf{x}}$, not merely if the old scorer is fooled.

4 The Manifold Representation Hypothesis

Concept erasure should remove a target attribute from a representation *without* damaging the unrelated structure that downstream computation relies on. While neural representations live in $\mathbb{R}^d$, learned representations produced by trained models do not fill this space uniformly. Instead, they concentrate near a much smaller region that we treat as a low-dimensional *manifold* embedded in the ambient space. This observation changes the geometry of erasure: an arbitrary direction in $\mathbb{R}^d$ may be a valid ambient vector, but it need not be tangent to the manifold of natural representations. Moving in such a direction pushes the representation *off the manifold*, into a region the encoder does not normally produce, where downstream computations behave unpredictably and unrelated information is distorted.

This is the central reason an off-manifold step damages *unknown* control concepts: most of what the representation encodes is unlabeled, but every encoded concept — labeled and unlabeled alike — is read out by downstream layers from the same manifold. Pushing off the manifold corrupts that shared substrate; staying on it bounds the damage by construction.

The hypothesis. We call this the *Manifold Representation Hypothesis* (MRH): the natural representations of a trained model concentrate on a low-dimensional manifold $\mathcal{M} \subset \mathbb{R}^d$, and an erasure step that moves each representation *along* $\mathcal{M}$ is structurally *surgical* — it edits exactly the degrees of freedom the model itself uses to encode information, while leaving the orthogonal ambient directions, which carry no learned signal, untouched. A step with off-manifold components, by contrast, damages the unlabeled control concepts that ride along the same manifold.

Linear vs. nonlinear erasure under MRH. Linear erasure has a closed-form, low-rank solution that is applied uniformly to every representation: the operator targets a small number of global moments (a mean direction, a covariance asymmetry) and cannot adapt per sample. Nonlinear erasure has no analogous closed form. Without one, we must estimate per-sample directions of concept presence (typically as the gradient of a learned scorer) and apply them iteratively. The estimate has no built-in constraint to remain on the manifold, so a per-sample step that locally reduces concept signal can also push the representation off-manifold.

A testable prediction. MRH makes a directly testable prediction: if the on-manifold constraint is what makes erasure surgical, then bolting it on top of *any* other erasure pipeline should reduce residual concept leakage without further damaging the controls. We confirm exactly this in §7.2, where stacking TanGCE on top of five prior erasers (LEACE, LEACE + CovMatch, INLP, IGBP, Obliviator) lowers nonlinear-probe leakage on every panel while preserving control accuracy.

5 Method: TanGCE family

TanGCE (*Tangent Gradient-based Concept Erasure*) operationalizes the Manifold Representation Hypothesis (§4) by constraining every erasure update to lie in the tangent space of $\mathcal{M}$ at the current point. At each round it trains a fresh nonlinear scorer on the current $(\mathbf{X}, \mathbf{y})$, projects the scorer's gradient onto a locally estimated tangent basis, and applies a spectrally-normalized trust-region update. The scorer (a 2-layer MLP with hidden width h=512, 800 steps) is intentionally distinct from the smaller evaluation probe (h=128, 200 SGD steps, §6.1), so that erasure quality is judged by an independently trained classifier rather than by the one that produced the update.

Local estimation of the manifold. A globally consistent estimate of $\mathcal{M}$ is intractable in the moderate-sample regime that erasure operates in (typically $n \approx 7000$ representations per setting in our experiments). We therefore estimate only the local tangent space $T_{\mathbf{x}_i}\mathcal{M}$ at each query point, using the local-PCA-of-secants construction of LTSA [Zhang and Zha, 2004]. This local view is well matched to concept erasure for two reasons. First, every smooth manifold is locally Euclidean, so the tangent estimate is asymptotically exact in a sufficiently small neighborhood and the standard linear-algebraic operations we need — gradient projection, trust-region step — transfer without modification. Second, internal representations encode a concept along different directions in different regions of $\mathcal{M}$: the axis that distinguishes the concept in one neighborhood is generally not the axis that distinguishes it elsewhere. Training a fresh nonlinear scorer at each iteration and projecting its gradient onto the locally estimated tangent basis lets us extract the locally relevant concept direction wherever a sample lies, and remove only its on-manifold component.

Limitation of the local estimate, and the closed-form fix. A first-order tangent estimate is only accurate within a small neighborhood. This makes TanGCE effective at removing the local nonlinear concept structure that nonlinear probes most readily exploit, but it is poorly suited to absorbing class-conditioned distributional moments that span the entire dataset — in particular the mean shift between the two classes, and the variance asymmetry along axes where the class-conditional second moments differ. These global moments are linear, not local: a closed-form affine eraser on the full distribution removes them in a single step with provably minimum representation damage. We therefore precompose TanGCE with two such closed-form preprocessing stages — LEACE [Belrose et al., 2023] for the first-moment alignment, and a rank-2 covariance-asymmetry projection (a rank-2 specialization of the k-LEACE family of Singh et al. 2024) for the second-moment residual — after which the iterative manifold loop refines what remains. We call these compositions TanGCE$^+$ (LEACE only) and TanGCE^{++} (LEACE + rank-2 CovMatch; we use "CovMatch" to refer to this rank-2 closed-form step throughout), retaining bare TanGCE (no preprocessing) as the variant that isolates the contribution of the manifold step. The closed-form stages are detailed in §5.1; the iterative manifold loop is detailed in §5.4–§5.4.

Manifold-aware erasure as a complementary refinement. Consistent with MRH (§4), the manifold step composes with *any* prior erasure technique: in §7.2 we show that running TanGCE on top of LEACE, LEACE + CovMatch, INLP, IGBP, or Obliviator further reduces nonlinear-probe leakage on every panel while preserving control accuracy. The gain is not specific to any one preprocessor, but to the fact that all of them leave behind the same locally curved residue that the manifold-constrained step removes.

5.1 Phase 1: closed-form moment preprocessing

Phase 1 is what turns TanGCE into TanGCE$^+$ (LEACE only) or TanGCE^{++} (LEACE + rank-2 CovMatch). It exhausts 1st- and 2nd-moment concept signal in closed form so that the subsequent manifold-aware loop only has to attack the residual nonlinear pocket.

LEACE (1st moment). Fit a LEACE eraser [Belrose et al., 2023] on $(\mathbf{X}_{\text{train}}, \mathbf{y})$ and apply to all splits. LEACE removes the rank-1 cross-covariance $\mathbf{\Sigma}_{xz}$, leaving no *linear* classifier above majority-class accuracy with provably minimum representation damage among affine projections.

Covariance-asymmetry projection (2nd moment). LEACE matches class means but leaves the within-class asymmetry

$$\Delta\mathbf{\Sigma} = \mathbf{\Sigma}_+ - \mathbf{\Sigma}_-. \tag{4}$$

Algorithm 1 TanGCE^{++}. Phase 1: LEACE + rank-2 covariance-asymmetry projection. Phase 2: iterative tangent-constrained step. Dropping line 1b yields TanGCE^{+}; dropping Phase 1 entirely yields TanGCE.

Require: Representations $\mathbf{X}$, concept labels $\mathbf{y}$, #rounds T, trust region ε, scorer-refit period K, scorer hidden h.

1: **Phase 1 – closed-form moment preprocessing** (Eqs. 4–6):
2: **1a.** $\mathbf{X} \leftarrow \mathrm{LEACE}_{\mathbf{y}}(\mathbf{X})$ (rank-1 mean projection)
3: **1b.** $\mathbf{X} \leftarrow \mathbf{X} - \mathbf{D}\mathbf{D}^{\top}\mathbf{X}$, with $\mathbf{D}$ from Eq. 5 (effective rank 2 post-LEACE)
4: Estimate tangent rank $r \leftarrow \max(8, \mathrm{TwoNN}(\mathbf{X}))$ (once, on the post-Phase-1 $\mathbf{X}$)
5: **Phase 2 – iterative tangent step** (Eqs. 9–15):
6: **for** $t = 1, \ldots, T$ **do**
7: **Refit scorer every** K **rounds.** If $t = 1$ or $(t-1) \bmod K = 0$: train a fresh 2-layer MLP scorer f_t (hidden h) on the current $(\mathbf{X}, \mathbf{y})$. Otherwise reuse f_{t-1}.
8: **Re-estimate the local tangent every round.** For each row $\mathbf{x}_i$, recompute the secant SVD on its k-NN cloud in the current $\mathbf{X}$ (Eq. 7); take the top-r right singular vectors as $\mathbf{B}_i$.
9: **For each row** $\mathbf{x}_i$: tangent-projected, spectrally-normalized trust-region step
10: $$\mathbf{x}_i \;\leftarrow\; \mathbf{x}_i - \lambda_i \sum_{k=1}^{r} \sigma_{i,k}\, a_{i,k}\, \mathbf{v}_i^{(k)}, \quad \|\Delta \mathbf{x}_i\|_2 \le \varepsilon \cdot s_i.$$
11: **end for**
12: **return** edited $\mathbf{X}$.

A nonlinear probe still recovers concept information from variance asymmetry along axes where $\mathbf{\Sigma}_+$ and $\mathbf{\Sigma}_-$ differ. We append a rank-2 projection: the top-2 eigenvectors of $\Delta\mathbf{\Sigma}$ ranked by $|\lambda|$, orthonormalized with a residual mean direction via QR,

$$\mathbf{D} = \mathrm{QR}\Big(\big[\,\hat{\mathbf{d}}_{\text{mean}} \;\big|\; \mathbf{e}_1, \mathbf{e}_2\,\big]\Big) \in \mathbb{R}^{d\times 3}, \tag{5}$$

$$\tilde{\mathbf{x}} = \mathbf{x} - \mathbf{D}\mathbf{D}^{\top}\mathbf{x}. \tag{6}$$

Because LEACE has aligned class means, $\hat{\mathbf{d}}_{\text{mean}} \approx \mathbf{0}$ and the effective rank removed is two. This is a special case of the k-LEACE framework of Singh et al. [2024] restricted to second-order eigenvectors, which are stable to estimate from $n \approx 7000$ examples while higher-order tensors are not. We fix $K{=}2$ across all 119 settings: for a binary concept, the top-2 eigenvectors of $\Delta\mathbf{\Sigma}$ carry the two axes of class-conditional variance asymmetry; subsequent eigenvectors describe noise rather than class-discriminative signal. Empirically, $K{\ge}3$ adds no leakage reduction and begins to erode control accuracy (see Tab. 4).

The two stages erase structurally distinct signal (mean shift vs. variance asymmetry) and together remove an effective rank of at most 3, negligible relative to the ambient dimension d across our settings ($d = 768$ for the CLIP ViT-B/32 image embedding, and $d \in [896, 5376]$ across the 13 LLM hidden sizes), exposing the residual nonlinear structure that Phase 2 attacks.

5.2 Local tangent estimation

We estimate $T_{\mathbf{x}_i}\mathcal{M}$ by local PCA/SVD of nearest-neighbor secants, following the local tangent-space approximation used in LTSA [Zhang and Zha, 2004]. For each query $\mathbf{x}_i$, let $\mathcal{N}_k(\mathbf{x}_i)$ be its k nearest neighbors. Stack the centered secants $\mathbf{s}_{j_m} = \mathbf{x}_{j_m} - \mathbf{x}_i$ row-wise into $\mathbf{S}_i \in \mathbb{R}^{k\times d}$ and take its SVD,

$$\mathbf{S}_i = \mathbf{U}_i \,\mathrm{diag}(\sigma_{i,1}, \ldots, \sigma_{i,k})\, \mathbf{V}_i^{\top}. \tag{7}$$

The right singular vectors $\mathbf{V}_i$ are the directions in which neighbors extend; the singular values $\sigma_{i,k}$ are how far they extend along each. The implied local covariance is $\mathbf{C}_i = \frac{1}{k}\mathbf{V}_i\,\mathrm{diag}(\sigma_{i,k}^2)\,\mathbf{V}_i^{\top}$. Truncating to the top r right singular vectors gives the tangent basis

$$\mathbf{B}_i = [\mathbf{v}_i^{(1)}, \ldots, \mathbf{v}_i^{(r)}] \in \mathbb{R}^{d\times r}, \tag{8}$$

where the tangent rank r is set automatically per setting by the TwoNN intrinsic-dimension estimator of Facco et al. [2017]. TwoNN reads off the local intrinsic dimension from the ratio of each point's distance to its nearest and second-nearest neighbor: under the assumption that the data lie on a

d^*-dimensional manifold with locally near-uniform density, those ratios follow a Pareto distribution whose shape parameter equals d^*, and TwoNN returns the maximum-likelihood fit. We need this because the tangent rank is dataset- and model-dependent — intrinsic dimension varies across LLM scales, families, and concepts, and across CLIP for vision — so a single fixed r would over- or under-truncate on most panels. We use TwoNN's per-setting estimate and floor it at $r \geq 8$ to guarantee enough degrees of freedom for the manifold-aware step. TwoNN is run once on the clean (post-Phase-1) representations and the resulting r is held fixed throughout the iterative loop; only the local secant SVD itself (the per-row $\mathbf{B}_i$) is recomputed every iteration on the current iterate's representations, so that the tangent basis tracks how the representation cloud reshapes under the edits. The tangent projector is

$$\mathbf{P}_{T_i} = \mathbf{B}_i \mathbf{B}_i^\top. \tag{9}$$

We treat the SVD basis as a first-order stand-in for $T_{\mathbf{x}_i}\mathcal{M}$, not as a guarantee of smooth-manifold structure.

5.3 Tangent-constrained update

The ambient unit gradient at $\mathbf{x}_i$ is $\mathbf{u}_i = \nabla f_t(\mathbf{x}_i)/\|\nabla f_t(\mathbf{x}_i)\|$. TanGCE replaces it with its tangent projection,

$$\mathbf{u}_i^{\tan} = \mathbf{P}_{T_i}\mathbf{u}_i = \mathbf{B}_i(\mathbf{B}_i^\top \mathbf{u}_i), \tag{10}$$

not re-normalized: $\|\mathbf{u}_i^{\tan}\| \leq 1$ acts as an overlap weight that attenuates the update when the ambient gradient has little support in the estimated tangent space. The isotropic update is

$$\tilde{\mathbf{x}}_i = \mathbf{x}_i - \lambda\,(\mathbf{x}_i^\top \mathbf{u}_i^{\tan})\,\mathbf{u}_i^{\tan}, \tag{11}$$

with λ a step-size scalar (replaced below by a closed-form per-sample λ_i). AmbGCE^{++} is the direct ambient ablation (used on both NLP and Vision): same closed-form preprocessing, same iterative loop, same scorer, same λ, but applies the update in the full ambient space without the manifold projection. We retain Eq. 11 as the conceptual entry point and as the $\alpha = 0$ ablation; the canonical TanGCE update is the spectrally-normalized generalization in §5.4.

5.4 Anisotropic trust-region step in tangent SVD coordinates

The simple update rule. The skeleton of every iteration is the tangent-projected gradient step of Eq. 11: subtract $\lambda \cdot (\mathbf{x}_i^\top \mathbf{u}_i^{\tan})\mathbf{u}_i^{\tan}$ from $\mathbf{x}_i$, where $\mathbf{u}_i^{\tan}$ is the tangent-projected unit gradient and λ is a step size. The remaining design questions are (i) how large λ should be per sample and (ii) along which tangent axes the step should put its mass. We never literally take a λ=1 step in practice; we treat it only as the bookkeeping reference at which the displacement vector is unscaled.

Local trust region as a constraint optimisation. A uniform global λ over-edits some samples while under-editing others. We instead solve, per sample, the one-dimensional constrained problem

$$\lambda_i \;=\; \arg\max_{\lambda \leq \lambda_{\max}} \lambda \quad \text{subject to} \quad \|\tilde{\mathbf{x}}_i - \mathbf{x}_i\|_2 \leq \varepsilon \cdot s_i, \tag{12}$$

i.e. the largest λ for which the displacement stays within a fraction ε of a per-sample local scale s_i (the precise definition of s_i comes below). Letting $\mathbf{v}_i$ be the unscaled tangent-step direction (the displacement vector that Eq. 11 would produce at the bookkeeping reference λ=1), the constraint $\|\lambda\,\mathbf{v}_i\|_2 \leq \varepsilon\, s_i$ has the closed-form maximiser

$$\boxed{\lambda_i = \min\left(\lambda_{\max},\ \frac{\varepsilon \cdot s_i}{\|\mathbf{v}_i\|}\right)} \tag{13}$$

with $\lambda_{\max} = 64$ a safety cap that bites only when $\|\mathbf{v}_i\| \to 0$. We use $\varepsilon = 0.1$ across every setting; λ_i then takes whatever (typically large) value the constraint admits, sample by sample.

Tangent axes have different "lengths". The local linear approximation produced by the secant SVD (Eq. 7) decomposes the tangent space into right singular vectors $\{\mathbf{v}_i^{(k)}\}$ and singular values $\{\sigma_{i,k}\}$. The singular values are the per-axis "lengths" — how far neighbors of $\mathbf{x}_i$ extend along each tangent direction. We will call the directions with large $\sigma_{i,k}$ the *long axes* of the neighborhood (high-σ for short) and the directions with small $\sigma_{i,k}$ the *thin axes* (low-σ). A direction-agnostic Euclidean radius treats every axis identically: it over-allocates step budget along thin axes (where the linear approximation is fragile and a small step in coordinate space already escapes the manifold) and under-allocates along long axes (where neighbors give the linear approximation generous support).

Per-axis budget via spectral normalisation. Respecting the per-axis lengths means measuring step magnitude in σ-normalised coordinates: divide each tangent coordinate of the move by the local singular value of its axis. This is Mahalanobis whitening expressed through the SVD — the local secant covariance is $\mathbf{C}_i = \frac{1}{k}\mathbf{V}_i \mathrm{diag}(\sigma_{i,k}^2)\mathbf{V}_i^\top$ and so $\mathbf{C}_i^{-1/2}$ acts diagonally as $\sigma_{i,k}^{-1}$ in the SVD basis, recovering the per-axis division for free. Concretely, we use the spectrally-normalised neighborhood radius

$$s_i = R_i^{\mathrm{spec}} \cdot s_i^{\mathrm{maha}}, \qquad R_i^{\mathrm{spec}} = \mathrm{median}_{j \in \mathcal{N}_k(i)} \|\mathbf{B}_i^\top(\mathbf{x}_j - \mathbf{x}_i)/\boldsymbol{\sigma}_i\|_2, \tag{14}$$

with $\boldsymbol{\sigma}_i = (\sigma_{i,1}, \ldots, \sigma_{i,r})$ and $s_i^{\mathrm{maha}} = 1/\|\boldsymbol{a}_i/(\boldsymbol{\sigma}_i + \delta)\|_2$ (component-wise), where $\boldsymbol{a}_i$ are the tangent coordinates of the displacement defined just below. ε is then a fraction of the local spectrally-normalised neighborhood; setting $s_i = r_i$ (mean k-NN distance) recovers the Euclidean radial scale and is reported in App. as an ablation.

The spectrally-normalised step itself replaces Eq. 11 with

$$\boxed{\tilde{\mathbf{x}}_i \;=\; \mathbf{x}_i \;-\; \lambda_i \sum_{k=1}^{r} \sigma_{i,k}^{\alpha}\, a_{i,k}\, \mathbf{v}_i^{(k)}} \tag{15}$$

where $a_{i,k} = \langle \mathbf{v}_i, \mathbf{v}_i^{(k)} \rangle$ are the SVD-basis coordinates of the unscaled tangent displacement and $\alpha = 1$ in the main experiments. Each tangent coefficient is now scaled by the local extent of the neighborhood along that axis: long-axis (high-σ) directions receive larger updates because the linear approximation is reliable there, and thin-axis (low-σ) directions are suppressed because moving along them most easily leaves the manifold.

Relation to local covariance and Mahalanobis scaling. The update can also be viewed as applying a positive spectral power of the local secant covariance. The empirical covariance of the centered secants has eigenvectors $\mathbf{v}_i^{(k)}$ and eigenvalues proportional to $\sigma_{i,k}^2$, so scaling each tangent coordinate by $\sigma_{i,k}$ corresponds, up to a global constant, to applying the positive square root of this local covariance ($\mathbf{C}_i^{1/2}$). This is the opposite of whitening: Mahalanobis normalisation downweights high-variance directions through an inverse covariance power, whereas we use a positive power to move preferentially along directions that are well supported by nearby representations. App. H details the full $\sigma_{i,k}^{\alpha}$ family with $\alpha \in \{0, 1, 2\}$, corresponding respectively to the isotropic ablation, our default, and the natural-gradient / full-Mahalanobis trust-region optimum.

The trust-region magnitude in Eq. 13 is enforced after spectral normalisation (replacing $\|\mathbf{v}_i\|$ by the norm of the spectrally-normalised displacement in the denominator), so the actual ambient move still respects $\|\tilde{\mathbf{x}}_i - \mathbf{x}_i\|_2 \leq \varepsilon \cdot s_i$. Anisotropy reshapes the *direction* of the step, not its budget.

5.5 Iterative loop and relation to baselines

Each round recomputes per-sample gradients and applies the manifold-aware update. The two refits inside the loop run on different cadences: the nonlinear scorer is retrained from scratch every K rounds (we use $K = 8$) on the current $(\mathbf{X}, \mathbf{y})$, while the local tangent basis is re-estimated every round, so the secant SVDs $\mathbf{B}_i$ track how the representation cloud reshapes after each edit even when the same scorer is reused. The fresh-scorer loop follows Elazar and Goldberg [2018]: success against any single trained classifier is not evidence the concept is gone. The iterative loop alone is TanGCE; combined with LEACE preprocessing it is TanGCE$^+$; combined with LEACE + rank-2 CovMatch it is TanGCE^{++}. AmbGCE^{++} keeps the same closed-form preprocessing, the same loop and the same trust region, but applies the update in the full ambient space — the direct ablation that isolates the manifold constraint. The two nonlinear baselines we compare against, IGBP [Iskander et al., 2023] and Obliviator [Akbari et al., 2025], share our supervision regime (target labels used during fitting, control concepts never seen by the eraser) and differ only in how the update is constructed: IGBP retracts toward an iteratively-retrained 2-class probe boundary; Obliviator solves an HSIC-penalised RKHS objective via random Fourier feature eigenvalue projection. TanGCE differs from both by constraining each update to a locally estimated tangent space.

6 Experimental setup

6.1 Metrics

Erasure precision S. A fresh 2-layer MLP probe (h=128, 200 SGD steps, patience 3) is trained on the intervened representations $\tilde{\mathbf{X}}$ and reports test accuracy $S = \mathrm{Acc}(h_{\mathrm{NL}}(\tilde{\mathbf{X}}), y)$. Lower is stronger erasure; the per-setting majority-class accuracy S_{floor} is the trivial lower bound. We report $S - S_{\text{floor}}$ in pp throughout.

Surgicality on control concepts ΔY. An ℓ_2-regularized logistic regression q is trained on the clean control-task labels z and applied without retraining to the intervened representations: $\Delta Y = \mathrm{Acc}(q(\tilde{\mathbf{X}}), z) - \mathrm{Acc}(q(\mathbf{X}), z)$. Smaller $|\Delta Y|$ is more surgical.

6.2 Settings, methods, and hyperparameters

Settings and probe. We evaluate on 119 settings: 39 NLP settings (13 LLM families $\times$ 3 concepts at the 50%-depth layer of each model) and 80 CelebA-CLIP settings (40 binary facial attributes $\times$ 2 control-set regimes). The 13 LLMs span 0.5B–27B parameters: Qwen2.5 (0.5B, 1.5B, 3B), Gemma-2 (2B, 9B, 27B), Llama-3.2 (1B, 3B), Mistral-7B-v0.1, Gemma-3 (1B, 4B, 12B, 27B). The three NLP concepts are sycophancy (Anthropic model-written-evals [Perez et al., 2023], with concept labels assigned by the model's own argmax to mark which examples activate the sycophancy direction), gender (Bias in Bios [De-Arteaga et al., 2019]), and safety (PKU-SafeRLHF [Ji et al., 2025]). Vision settings use CLIP ViT-B/32 [Radford et al., 2021] pooled embeddings of CelebA [Liu et al., 2015]; per concept, the 5 least-correlated and 5 most-correlated other attributes form the two control sets. Concept removal is judged by retraining a fresh nonlinear probe (2-layer MLP, hidden $h = 128$, 200 SGD steps, patience 3) on the edited representations.

Compared methods. The TanGCE family contributes three rows: TanGCE (manifold-aware iterative loop alone, isolating the contribution of the manifold projection); TanGCE^{+} (LEACE preprocessing + manifold loop); TanGCE^{++} (LEACE + rank-2 CovMatch preprocessing + manifold loop, the headline pipeline; "CovMatch" is shorthand for the rank-2 covariance-asymmetry projection defined in §5.1). AmbGCE^{++}: same closed-form preprocessing, same loop and trust region as TanGCE^{++} but applies the update in the full ambient space, the direct ablation isolating the manifold constraint. The same recipe is used on both NLP and Vision. LEACE [Belrose et al., 2023] and LEACE + CovMatch: closed-form one-shot linear and rank-2 covariance erasers, the preprocessing stages of TanGCE^{+} / TanGCE^{++} reported alone. INLP [Ravfogel et al., 2020]: iterative linear nullspace projection, with a fresh nonlinear probe re-evaluated after each rank-removal step. IGBP [Iskander et al., 2023]: iterative 2-class probe with Newton-step boundary projection, also re-evaluated each round under the same protocol. Obliviator [Akbari et al., 2025]: the strongest published nonlinear baseline (HSIC encoder + RFF-EVP projection, paper hyperparameters); supervised on the target concept and unsupervised on the controls, the same regime as our method. λ-based methods report at $\lambda = 29.31$, the mean per-row λ_i that TanGCE^{++} produces (App. Tab. 11), giving them TanGCE^{++}'s effective step magnitude. All methods are run under the trajectory-probe protocol, with a fresh nonlinear MLP scoring S and Y after every projection step.

CelebA control-set construction. On Vision, each of the 40 CelebA attributes is in turn used as the erasure target; the controls for that target are 5 of the remaining 39 attributes, and we report two complementary regimes. Let $\mathbf{R}_{k\ell}$ be the empirical Pearson correlation between attribute labels k and ℓ on the train split. For each target k, the *5 least-correlated* controls are the bottom 5 of the remaining 39 by absolute correlation $|\mathbf{R}_{k\ell}|$ (low-entanglement regime, the standard surgicality choice in prior work), and the *5 most-correlated* are the top 5 (high-entanglement regime, the surgicality stress test). For example, when erasing *Male* the most-correlated controls are *Wearing_Lipstick* ($|r| \approx 0.80$), *Heavy_Makeup*, *No_Beard*, *Attractive*, *5_o_Clock_Shadow* — attributes the dataset labels as nearly redundant with gender. Together this yields 80 settings (40 targets $\times$ 2 regimes); the per-target listings of both regimes with their $|r|$ values are in App. Tabs. 9–10, and the construction details (formula, edge cases) are in App. F.

CelebA evaluation protocol. For each target $\times$ regime, we extract pooled CLIP ViT-B/32 image embeddings of CelebA on the official train/val/test splits, fit the eraser on train, and re-evaluate after

every projection step on test. The fresh nonlinear MLP (above) measures target leakage S on the target attribute. Control-task accuracy Y is the mean over the regime's 5 controls of an independent fresh MLP trained on each control attribute; the control-damage signal is $\Delta Y = Y - Y_{\text{clean}}$ (negative when controls degrade). The four budgets $\Delta Y \in \{-1, -3, -5, -10\}$pp are applied to this mean drop, so a method is in budget on a setting when its average control accuracy across the 5 selected controls dropped by at most B percentage points relative to the clean (un-erased) representation.

Hyperparameters. The per-sample trust-region λ_i (Eq. 13) eliminates the global λ from the proposed method. We fix step-controlling hyperparameters across all 119 settings: $\varepsilon = 0.1$, scorer hidden $h = 512$, scorer training steps $= 800$, scorer-refit every 8 outer iterations, $T = 60$ outer iterations, TwoNN intrinsic-dimension floor $k_{\min} = 8$, safety cap $\lambda_{\max} = 64$, Phase-1 projection rank $K = 2$. Every method in Tab. 1 runs at fixed hyperparameters with no per-setting tuning.

7 Results

7.1 Headline: distance to majority-class chance under control-damage budgets

Tab. 1 reports the headline number for each method at four control-damage budgets, anchored to the per-setting majority-class floor. The TanGCE family contributes three rows — TanGCE (manifold loop only), TanGCE$^+$ (LEACE + manifold), TanGCE^{++} (LEACE + CovMatch + manifold) — so the contribution of each closed-form preprocessing stage is visible alongside the headline. For each setting and budget $B \in \{1, 3, 5, 10\}$pp we extract the trajectory step closest to floor (smallest $|S - S_{\text{floor}}|$) subject to $|Y_{\text{clean}} - Y| \leq B/100$ and report the signed $(S - S_{\text{floor}})$ in pp averaged across the 39 settings. The right sub-column of each block reports settings driven to chance ($|S - S_{\text{floor}}| \leq 0.5$pp) over settings reaching the budget. Bold marks every method whose mean is within 0.05pp of the column-best. INLP and IGBP run under the same trajectory-probe protocol: each round projects out one source-probe direction, and a fresh nonlinear MLP probe (h=128, 200 SGD, patience 3) re-evaluates S and Y on the post-projection state.

How to read the @chance counts. The denominator of the @chance sub-column is *not* the full setting count: it is the number of settings on which the method produced at least one trajectory step inside the control-damage budget. A method that overshoots the budget on its very first step on some settings simply never appears in the running for those settings – it neither contributes to the mean nor counts toward the @chance numerator. This matters most in the high-entanglement CelebA regime (§7.4), where a method can post a flattering mean of $+0.0$pp by qualifying on only the easy settings and being silently absent from the hard ones. We therefore read each cell as the pair (mean leakage among reachable settings, @chance / reachable settings), and treat a method as fully admissible only when its denominator is close to the full setting count.

TanGCE^{++} is the only method whose mean approaches chance at every budget: $+2.6 \rightarrow +0.4$pp from ≤ 1pp to ≤ 10pp. Obliviator plateaus at $+4.0$ then $+2.7$pp. AmbGCE^{++} (same closed-form preprocessing and trust region as TanGCE^{++}, but no manifold constraint) leaves $+7$ to $+10$pp and fails to reach the budget on 6 to 15 of 39 settings. The two iterative source-probe baselines plateau strictly above the trust-region methods: IGBP at $+9.4$pp and INLP at $+13.6$pp at the loosest budget, with @chance counts (4–8/39) below LEACE + CovMatch's $16/39$. The pattern follows the geometric reading: cumulative linear nullspace projection (INLP) and Newton-step projection on a 2-class probe (IGBP) close the source axis but do not constrain the step to remain on the local manifold, so a fresh nonlinear probe still recovers a residual nonlinear pocket. LEACE alone leaves an ≈ 18pp gap that closed-form rank-2 covariance matching halves. The two ablation rows in the TanGCE family pin down the contribution of each closed-form preprocessing stage: pure TanGCE (no preprocessing) leaves mean leakage in the $+6.7$ to $+9.9$pp range and reaches @chance on only 9–$11/39$ settings; TanGCE$^+$ (LEACE preprocessing) collapses leakage to $+0.6$–$+2.8$pp and lifts @chance to 18–$26/39$; TanGCE^{++} (LEACE + CovMatch) reaches chance on 26–$33/39$ with $+0.4$–$+2.6$pp leakage, while the best baseline (Obliviator) reaches chance on 13–$17/39$. The LEACE stage carries most of the gain — removing the global mean shift gives the local manifold loop a tractable residual — and the rank-2 CovMatch step then converts that into a higher fraction of fully-erased settings (especially at the loosest budget, $33/39$ vs. $26/39$). The full per-setting breakdown is in App. Tab. 4.

Table 1: **Main result.** *Leakage (pp)*: signed $S - S_{\text{floor}}$ at the trajectory step closest to majority-class chance, in pp; 0 = at chance, lower in absolute value = stronger erasure. *@chance*: settings driven to chance ($|S - S_{\text{floor}}| \leq 0.5$pp) over settings the method reached the budget on. All 39 NLP panels (13 model families $\times$ 3 concepts) at 50%-depth, fresh nonlinear probe (h=128, 200 SGD, patience 3); LEACE-only and LEACE+CovMatch (rank-2 covariance-asymmetry projection) are one-shot closed-form. TanGCE^{++} is the proposed pipeline at fixed ε=0.1, h=512; TanGCE drops both preprocessing stages and TanGCE^{+} drops only CovMatch. Bold = column-best; multiple methods within 0.05pp of the best are all bolded. Per-panel detail across all 39 panels is in App. Tab. 4.

Method	$\Delta Y \leq$ 1pp		$\Delta Y \leq$ 3pp		$\Delta Y \leq$ 5pp		$\Delta Y \leq$ 10pp	
	Leakage	@chance↑	Leakage	@chance	Leakage	@chance	Leakage	@chance
LEACE	+18.3	5/38	+18.6	5/39	+18.6	5/39	+18.6	5/39
LEACE+CovMatch	+9.6	12/30	+8.9	16/38	+8.9	16/38	+8.9	16/39
INLP	+14.1	7/38	+13.8	7/39	+13.7	8/39	+13.6	8/39
IGBP	+9.5	4/37	+9.4	4/38	+9.4	4/38	+9.4	4/38
Obliviator	+4.3	13/39	+4.0	13/39	+4.0	13/39	+2.7	17/39
AmbGCE^{++}	+7.3	8/24	+9.3	10/31	+9.7	12/32	+9.5	13/33
TanGCE	+9.9	9/37	+8.3	9/38	+7.8	10/39	+6.7	11/39
TanGCE^{+}	+2.8	18/39	**+0.8**	24/39	**+0.6**	25/39	+0.6	26/39
TanGCE^{++}	**+2.6**	26/39	+1.3	29/39	**+0.6**	33/39	**+0.4**	33/39

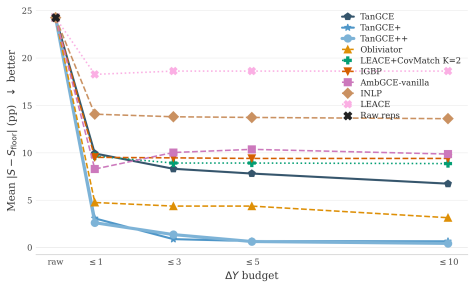


Figure 2: **Aggregated NLP leakage vs. control-damage budget.** Mean $|S - S_{\text{floor}}|$ in pp across the 39 NLP settings (13 model families $\times$ 3 concepts at 50%-depth) for each method, plotted against the four control-damage budgets $\Delta Y \in \{1, 3, 5, 10\}$pp. The black "X" at $x = 0$ is the shared starting point: mean nonlinear-probe leakage on the raw, un-erased representations ($\sim$24pp); every line begins from this anchor. Lower is better; the y-axis floor is the per-setting majority-class chance. The TanGCE family (TanGCE^{++}, TanGCE^{+}, TanGCE) traces the lowest envelope at every budget — TanGCE^{++} below 1pp from $\leq$ 3pp onward — while baselines (Obliviator, LEACE + CovMatch, IGBP, AmbGCE, INLP, LEACE) plateau at 4–18pp regardless of how much utility one is willing to sacrifice. The visual gap between the TanGCE curves and the rest reproduces the column-best ranking in Tab. 1.

7.2 TanGCE complements every Phase-1 baseline

To isolate the marginal value of the manifold-aware Phase 2 (TanGCE), we apply the canonical TanGCE recipe (ε=0.1, h=512, scorer warm-start, $\leq$ 60 outer iterations) on top of each baseline's converged representations and re-evaluate under the same ΔY budget convention as Tab. 1 (control-damage measured against the original clean reps' Y). Tab. 2 reports the result. TanGCE reduces leakage on every Phase-1 baseline that lands in budget — LEACE drops from $\sim$20pp to $<$1pp, INLP from $\sim$16pp to $\sim$2pp, IGBP from $\sim$11pp to $\sim$1pp — without further degrading Y (the post-TanGCE step is selected from the same in-budget slice as the Phase-1-alone reading). Obliviator is a special case: it overshoots ΔY on every sycophancy and gender setting (26/26), so it is in budget only on the 13 safety settings, where its encoder collapses Y to the panel's Y_{floor} and the leakage shows 0.0pp as a vacuous artefact rather than a surgical erasure (marked † throughout the row).

The TanGCE^{++} reference row is the SOTA pipeline (LEACE + CovMatch + manifold), reproduced here for context. The TanGCE row is the manifold loop on raw reps with no Phase 1 — already competitive on leakage (mean 7.7pp at $\leq$ 5pp), but with much higher reach instability than any LEACE-stacked variant. The pattern is consistent with the geometric reading: every Phase-1 step

Table 2: **TanGCE complements every Phase-1 baseline.** Mean $|S - S_{\text{floor}}|$ in pp at the trajectory step closest to chance, with ΔY measured against the original clean reps (same budget convention as Tab. 1). For each baseline B, the "alone" column reports B's endpoint leakage on settings where B kept Y within the budget; the "Improvement" column is the mean reduction in leakage after stacking TanGCE on B's reps (positive = TanGCE moved closer to chance), evaluated on the same settings. For the reference rows (TanGCE on raw reps and the SOTA pipeline LEACE + CovMatch $\rightarrow$ TanGCE) there is no Phase-1-alone baseline; the Improvement slot italicises the absolute post-TanGCE leakage instead. Reach = (settings in budget)/(settings with data). The TanGCE-only and LEACE + CovMatch $\rightarrow$ TanGCE rows are reference rows for context (TanGCE on clean reps; full SOTA pipeline). †**Obliviator surgicality warning:** Obliviator overshoots the ΔY budget on *every* sycophancy and gender setting ($26/26$); the only settings on which it lands in budget are the 13 safety settings, where the safety task accuracy floor happens to coincide with Obliviator's encoder-collapse Y. The 0.0pp leakage on those 13 settings reflects this collapse, not a genuine surgical erasure — it is an artefact of Obliviator destroying control accuracy until Y matches the panel's Y_{floor}.

Method	$\Delta Y \leq 1$pp		$\Delta Y \leq 3$pp		$\Delta Y \leq 5$pp		$\Delta Y \leq 10$pp	
	alone	Improvement	alone	Improvement	alone	Improvement	alone	Improvement
TanGCE (no Phase 1)	–	*9.2* (37/39)	–	*8.4* (39/39)	–	*7.7* (39/39)	–	*6.8* (39/39)
LEACE $\rightarrow$ TanGCE	19.1 (38/39)	$\mathbf{+16.9}$	18.9 (39/39)	$\mathbf{+18.2}$	18.9 (39/39)	$\mathbf{+18.3}$	18.9 (39/39)	$\mathbf{+18.3}$
INLP $\rightarrow$ TanGCE	15.2 (36/39)	$\mathbf{+12.9}$	16.1 (38/39)	$\mathbf{+14.1}$	16.0 (39/39)	$\mathbf{+14.1}$	16.0 (39/39)	$\mathbf{+14.2}$
IGBP $\rightarrow$ TanGCE	11.3 (37/39)	$\mathbf{+9.4}$	11.2 (39/39)	$\mathbf{+9.7}$	11.2 (39/39)	$\mathbf{+10.1}$	11.2 (39/39)	$\mathbf{+10.3}$
Obliviator $\rightarrow$ TanGCE	$0.0^{\dagger}$ (13/39)	$\mathbf{+0.0}^{\dagger}$	$0.0^{\dagger}$ (13/39)	$\mathbf{+0.0}^{\dagger}$	$0.0^{\dagger}$ (13/39)	$\mathbf{+0.0}^{\dagger}$	$0.0^{\dagger}$ (18/39)	$\mathbf{+0.0}^{\dagger}$
LEACE + CovMatch $\rightarrow$ TanGCE	–	*2.9* (33/39)	–	*1.2* (38/39)	–	*0.7* (39/39)	–	*0.4* (39/39)

that closes a linear or second-moment leakage axis frees TanGCE's per-row trust region to attack the residual nonlinear pocket, and the manifold-aware step erases that pocket without escaping the in-budget Y slice.

7.3 Per-model bars and the surgicality signal

The gain is largest on gender (Fig. 3), where the rank-2 covariance projection of Phase 1 is most informative because gender concept information loads on second-moment asymmetry that LEACE alone cannot reach. The pattern is the geometric content of the manifold-awareness claim: TanGCE^{++} reaches chance while damaging Y less than the alternatives. Phase 1 alone (LEACE + CovMatch) and the manifold-aware Phase 2 (TanGCE) contribute structurally distinct erasure: closed-form moment matching exhausts what global affine projections can reach, and the per-row tangent step erodes the residual nonlinear pocket. App. Tab. 3 reports the Phase-1-only ablation across all 39 settings.

7.4 Cross-modal generalization: CelebA dual control regimes

Setup. CelebA evaluation uses the same fixed-$\varepsilon = 0.1$ recipe as the NLP grid; the dual control-set construction (5 least- vs. 5 most-correlated attributes per target) and the ΔY control-damage budget protocol are detailed in §6.2, with the per-target control listings in App. Tabs. 9–10.

Fig. 4 aggregates the 40 attributes of the least-correlated regime into one mean-leakage bar per method at the two tightest budgets, with the number of attributes each method produced an in-budget result on annotated on the bar. The high-entanglement (most-correlated) regime, the per-attribute breakdown, and the wider per-method/per-budget summary are deferred to the appendix — App. Tab. 5 (the full per-method $\times$ per-budget summary across both regimes), App. Fig. 8 (per-attribute bar chart at all four budgets), and App. Tabs. 6–8 (numerical per-attribute breakdown at $\Delta Y \in \{1, 3, 5\}$pp).

The headline pattern, read off Fig. 4 and App. Tab. 5, is the count of attributes for which each method produced an in-budget leakage value at each of the four budgets:

- **Least-correlated regime**: Obliviator presents results on $15, 29, 39, 40$ of 40 attributes at $\Delta Y \leq \{1, 3, 5, 10\}$pp; TanGCE^{++} presents results on $39, 40, 40, 40$. The bar chart in Fig. 4 surfaces the two tightest budgets.
- **Most-correlated regime**: Obliviator presents results on $2, 15, 30, 34$ of 40; TanGCE^{++} presents results on $16, 34, 38, 40$. The gap is largest on the entangled targets (*Male*/most: every Obliv step $\in [-21, -13]$pp; *Heavy_Makeup*, *Wearing_Lipstick*, *Smiling* all walkovers).

Figure 3: **Gender (Bias in Bios): per-model nonlinear-probe accuracy at the tightest and a moderate control-damage budget.** Two budgets are shown ($\Delta Y \leq 1$pp and ≤ 5pp); each model group has three bars (gray = clean, blue = TanGCE^{++}, red = Obliviator) plus a black tick at the per-model majority-class floor. Across all 13 models and both budgets, TanGCE^{++} drives the probe to or below the floor; Obliviator stays close to the clean-rep accuracy on most settings. The full four-budget version of this figure is in App. Fig. 5; sycophancy and safety analogues are in App. Fig. 6–7.

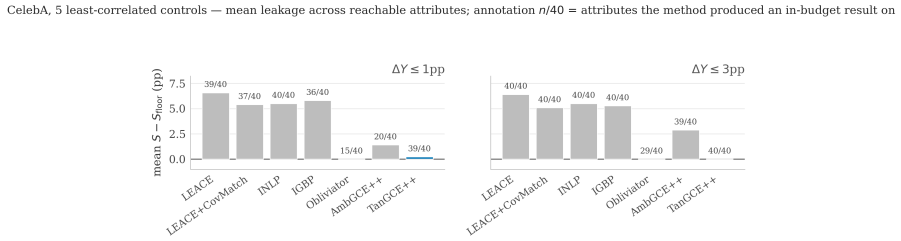


Figure 4: **CelebA aggregated leakage, 5 least-correlated controls.** Mean $S - S_{\text{floor}}$ in pp across the attributes a method produced an in-budget result on, for each method at the two tightest control-damage budgets ($\Delta Y \leq 1$pp and ≤ 3pp); lower is better, 0 is at chance. The "$n/40$" annotation on each bar is exactly that count — the number of CelebA attributes for which we could report a leakage value at this budget. TanGCE^{++} (blue) presents results on 39 of 40 attributes at ≤ 1pp and 40 of 40 at ≤ 3pp, with mean leakage essentially at chance; Obliviator (orange) presents results on only 15 and 29 of 40 at the same budgets, and its 0.0pp mean is averaged over that smaller subset.

On the attributes where both methods land, mean leakage is essentially indistinguishable: in the least-correlated regime TanGCE^{++} and Obliviator both sit at ~ 0pp; in the most-correlated regime TanGCE^{++} sits at $+0.4$–$+2.6$pp depending on the budget while Obliviator's 0.0 is computed only over the small subset it could enter. Linear and source-probe baselines (LEACE, LEACE + CovMatch, INLP, IGBP) plateau at $+5$ to $+6$pp gap above chance at every budget regardless of regime. AmbGCE^{++} is the most unstable trust-region method: in the most-correlated regime only 1 of 40 attributes reaches the budget at ≤ 1pp, and on the settings it does land it sits $+2$ to $+7$pp above chance. The geometric reading: cumulative linear nullspace projection (INLP) and Newton-step projection on a 2-class probe (IGBP) close the source axis but do not constrain the step to remain on the local manifold; AmbGCE^{++} has the same budget but escapes the manifold; only the on-manifold step erodes the residual nonlinear pocket while keeping the edit inside the per-attribute Y budget.

Pipeline note. The CelebA TanGCE^{++} and matched ambient AmbGCE^{++} runs use the same Phase 1 as the NLP headline (LEACE + rank-2 CovMatch).

8 Discussion

Two coupled mechanisms produce the surgicality gain. The headline pattern is consistent: the only method that matches or beats Obliviator on probe accuracy at every budget while damaging the control task less is the tangent-projected one (Tab. 4, Fig. 3). Two coupled mechanisms produce this pattern. The tangent constraint defines *where* edits live: ambient updates carry components weakly aligned with the local representation support, and tangent projection removes them at first order. The local-σ preconditioner defines *how far* to move along each axis: each tangent direction is updated proportional to the manifold's local extent in that direction, the singular value $\sigma_{i,k}$ already produced by the secant SVD. Together they restrict every step to a fraction of the local manifold-supported neighborhood. The signature is most visible in the most-correlated CelebA regime, where the dominant tangent axes are shared between target and controls and the preconditioned step is the only one that lands inside the budget on every setting.

Why the gain is largest on gender. Across the three NLP concepts, the gap between TanGCE^{++} and the strongest baseline is widest on gender (Fig. 3, Tab. 4). Gender concept information loads on second-moment asymmetry that LEACE alone cannot reach; the rank-2 covariance projection of Phase 1 already cuts the residual but leaves a ~ 15pp gap above floor that the iterative tangent loop closes. Sycophancy is the easy concept: Phase 1 alone is indistinguishable from the floor (App. Tab. 3), so the iterative loop adds nothing. Safety sits between the two. The per-concept ordering is consistent across model families and scales (0.5B–27B parameters); the tangent advantage is therefore not architecture-specific but tracks the underlying nonlinear structure of the concept.

Connection to low-rank fine-tuning. The local low-rank subspace structure that the tangent estimator exploits has a parameter-space analogue. Aghajanyan et al. [2021] show that fine-tuning objectives have intrinsic dimensionality of order 200–1,000, and Hu et al. [2022] exploit this directly via low-rank weight updates. Our finding is the activation-space analogue of the same phenomenon: hidden-state directions that carry task-specific information occupy a much smaller local support than the full ambient space. The two are mechanistically linked: if layer-ℓ activations have intrinsic rank m, then weight-space gradients at that layer are also rank-m at the activation level, formed as outer products of the low-rank activations and the corresponding error signals. From this perspective, LoRA and tangent-constrained concept erasure both follow from the same underlying structure in large-model representations, and the operative distinction is whether the update is allowed to act freely in ambient space or is constrained to the manifold support that the representations actually occupy.

9 Limitations

Single-seed snapshot. The headline tables report a single deterministic seed=0 run on the 39 NLP settings and the 80 CelebA-CLIP settings. Multi-seed variance across the 39 NLP settings will be reported in App. as the additional runs complete; the seed=0 numbers are not preregistered as the final headline values, only the result available at submission.

Measurement scope. Concept removal is judged by retraining a fresh 2-layer MLP probe (h=128, 200 SGD steps, patience 3) on the edited representations. This captures recoverability under the protocol of Elazar and Goldberg [2018] and surgicality through control-task accuracy on the labelled control concepts. It does not by itself establish broad capability preservation across unrelated tasks: a corpus-level evaluation on held-out general-domain text is left to future work. The cross-modal evidence comes from a single CLIP encoder (ViT-B/32 on CelebA); evaluations against other vision encoders or domains are out of scope here.

Geometry as supporting evidence, not causal proof. The local manifold motivation is operational, not a theorem. TwoNN intrinsic-dimension estimates and secant-SVD bases are first-order approximations of an unobserved tangent space. We do not connect tangent-estimation accuracy to

erasure quality through a closed-form bound, and we do not claim that off-support displacement is the sole mechanism behind every leakage–utility tradeoff. The geometry diagnostics support tangent projection as a useful inductive bias rather than as a complete causal account.

10 Conclusion

We introduced the TanGCE family, a nonlinear concept erasure approach motivated by a local manifold view of representation space. Hidden states live in $\mathbb{R}^d$ but natural representations concentrate near a much smaller, structured support; tangent projection plus a local-σ preconditioner restricts every erasure step to a fraction of the locally supported neighborhood. TanGCE (the manifold-aware iterative loop alone), TanGCE$^+$ (LEACE + manifold), and TanGCE^{++} (LEACE + rank-2 covariance + manifold) compose this manifold-aware step with closed-form preprocessing, exposing the contribution of each preprocessing stage.

Across 119 settings under fixed method hyperparameters, TanGCE^{++} reaches the per-setting majority-class probe-accuracy floor on more settings than the strongest published baseline while damaging the control task less. The gain is largest on the most nonlinear concept (gender) and, on CelebA, in the most-correlated control regime; both are regimes where ambient updates most readily push representations off the manifold. We treat the result as evidence that manifold-constrained erasure improves surgicality, not as a closed-form bound.

Several directions remain open. First, extending the TanGCE family to multi-layer erasure, where the tangent basis must be re-estimated after each layer's intervention, is a natural next step. Second, our tangent estimator is a first-order PCA-based stand-in; richer local manifold estimators may yield tighter tangent bases on the settings where the SVD truncation is the bottleneck. Third, connecting tangent-estimation accuracy to erasure quality through a formal bound would replace the empirical evidence we report here with a guarantee.

References

Armen Aghajanyan, Sonal Gupta, and Luke Zettlemoyer. Intrinsic dimensionality explains the effectiveness of language model fine-tuning. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 7319–7328. Association for Computational Linguistics, 2021. doi: 10.18653/v1/2021.acl-long.568. URL `https://aclanthology.org/2021.acl-long.568`.

Ramin Akbari, Milad Afshari, and Vishnu Naresh Boddeti. Obliviator reveals the cost of nonlinear guardedness in concept erasure. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. URL `https://neurips.cc/virtual/2025/poster/118956`.

Matan Avitan, Ryan Cotterell, Yoav Goldberg, and Shauli Ravfogel. A practical method for generating string counterfactuals. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 3267–3286. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.findings-naacl.180. URL `https://aclanthology.org/2025.findings-naacl.180/`.

Nora Belrose, David Schneider-Joseph, Shauli Ravfogel, Ryan Cotterell, Edward Raff, and Stella Biderman. Leace: Perfect linear concept erasure in closed form. In *Advances in Neural Information Processing Systems 36*, 2023. URL `https://papers.nips.cc/paper_files/paper/2023/hash/d066d21c619d0a78c5b557fa3291a8f4-Abstract-Conference.html`.

Maria De-Arteaga, Alexey Romanov, Hanna Wallach, Jennifer Chayes, Christian Borgs, Alexandra Chouldechova, Sahin Geyik, Krishnaram Kenthapadi, and Adam Tauman Kalai. Bias in bios: A case study of semantic representation bias in a high-stakes setting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency*, FAT* '19, pages 120–128. ACM, 2019. doi: 10.1145/3287560.3287572.

Yanai Elazar and Yoav Goldberg. Adversarial removal of demographic attributes from text data. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 11–21, Brussels, Belgium, 2018. Association for Computational Linguistics. doi: 10.18653/v1/D18-1002. URL `https://aclanthology.org/D18-1002/`.

Yanai Elazar, Shauli Ravfogel, Alon Jacovi, and Yoav Goldberg. Amnesic probing: Behavioral explanation with amnesic counterfactuals. *Transactions of the Association for Computational Linguistics*, 9:160–175, 2021. doi: 10.1162/tacl_a_00359. URL https://aclanthology.org/2021.tacl-1.10/.

Elena Facco, Maria d'Errico, Alex Rodriguez, and Alessandro Laio. Estimating the intrinsic dimension of datasets by a minimal neighborhood information. *Scientific Reports*, 7:12140, 2017. doi: 10.1038/s41598-017-11873-y.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022. URL https://openreview.net/forum?id=nZeVKeeFYf9.

Shadi Iskander, Kira Radinsky, and Yonatan Belinkov. Shielded representations: Protecting sensitive attributes through iterative gradient-based projection. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 5961–5977, Toronto, Canada, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-acl.369. URL https://aclanthology.org/2023.findings-acl.369/.

Jiaming Ji, Donghai Hong, Borong Zhang, Boyuan Chen, Juntao Dai, Boren Zheng, Tianyi Qiu, Jiayi Zhou, Kaile Wang, Boxuan Li, Sirui Han, Yike Guo, and Yaodong Yang. PKU-SafeRLHF: Towards multi-level safety alignment for LLMs with human preference. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Main Conference)*. Association for Computational Linguistics, 2025. URL https://arxiv.org/abs/2406.15513. arXiv:2406.15513.

Ziwei Liu, Ping Luo, Xiaogang Wang, and Xiaoou Tang. Deep learning face attributes in the wild. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 3730–3738, 2015. doi: 10.1109/ICCV.2015.425.

Ethan Perez, Sam Ringer, Kamilė Lukošiūtė, Karina Nguyen, Edwin Chen, Scott Heiner, Craig Pettit, Catherine Olsson, Sandipan Kundu, Saurav Kadavath, Andy Jones, Anna Chen, Benjamin Mann, Brian Israel, Bryan Seethor, Cameron McKinnon, Christopher Olah, Da Yan, Daniela Amodei, Dario Amodei, Dawn Drain, Dustin Li, Eli Tran-Johnson, Guro Khundadze, Jackson Kernion, James Landis, Jamie Kerr, Jared Mueller, Jeeyoon Hyun, Joshua Landau, Kamal Ndousse, Landon Goldberg, Liane Lovitt, Martin Lucas, Michael Sellitto, Miranda Zhang, Neerav Kingsland, Nelson Elhage, Nicholas Joseph, Noemi Mercado, Nova DasSarma, Oliver Rausch, Robin Larson, Sam McCandlish, Scott Johnston, Shauna Kravec, Sheer El Showk, Tamera Lanham, Timothy Telleen-Lawton, Tom Brown, Tom Henighan, Tristan Hume, Yuntao Bai, Zac Hatfield-Dodds, Jack Clark, Samuel R. Bowman, Amanda Askell, Roger Grosse, Danny Hernandez, Deep Ganguli, Evan Hubinger, Nicholas Schiefer, and Jared Kaplan. Discovering language model behaviors with model-written evaluations. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 13387–13434, Toronto, Canada, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-acl.847. URL https://aclanthology.org/2023.findings-acl.847.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 8748–8763. PMLR, 2021. URL https://proceedings.mlr.press/v139/radford21a.html.

Shauli Ravfogel, Yanai Elazar, Hila Gonen, Michael Twiton, and Yoav Goldberg. Null it out: Guarding protected attributes by iterative nullspace projection. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 7237–7256, Online, 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.acl-main.647. URL https://aclanthology.org/2020.acl-main.647/.

Shauli Ravfogel, Francisco Vargas, Yoav Goldberg, and Ryan Cotterell. Adversarial concept erasure in kernel space. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language*

Processing, pages 6034–6055. Association for Computational Linguistics, 2022. doi: 10.18653/v1/2022.emnlp-main.405. URL https://aclanthology.org/2022.emnlp-main.405/.

Shashwat Singh, Shauli Ravfogel, Jonathan Herzig, Roee Aharoni, Ryan Cotterell, and Ponnurangam Kumaraguru. Representation surgery: Theory and practice of affine steering. In *Proceedings of the 41st International Conference on Machine Learning*, Proceedings of Machine Learning Research. PMLR, 2024. URL https://arxiv.org/abs/2402.09631. arXiv:2402.09631.

Loring W. Tu. *An Introduction to Manifolds*. Universitext. Springer, 2 edition, 2011. doi: 10.1007/978-1-4419-7400-6. URL https://luis.impa.br/aulas/anvar/Tu_AnIntroductionToManifolds.pdf.

Zhenyue Zhang and Hongyuan Zha. Principal manifolds and nonlinear dimensionality reduction via tangent space alignment. *SIAM journal on scientific computing*, 26(1):313–338, 2004.

A Phase-1 closed-form ablation

Tab. 3 reports a fresh nonlinear probe (h=128, 200 SGD steps, patience 3) trained on three representation states per setting: clean, post-LEACE, post-(LEACE + CovMatch). Sycophancy is fully exhausted by Phase 1 (mean NL 0.563 indistinguishable from floor 0.560 and from TanGCE^{++}'s 0.560). Gender is not: Phase 1 leaves 14.8pp residual above floor that the manifold-aware Phase 2 closes to 1.0pp. Safety follows the gender pattern with $11.5 \rightarrow 0.2$pp.

B Per-model bar charts (sycophancy, safety, and full-budget gender)

Sycophancy and safety analogues of the gender per-model bar chart in the main text (Fig. 3), plus the full four-budget version of the main-text gender figure. On sycophancy, both TanGCE^{++} and Obliviator are at floor on essentially every model and budget; sycophancy is the easy concept. On safety, TanGCE^{++} sits at or below the per-model floor on all 13 settings and all four budgets, while Obliviator overshoots floor by 1–3pp on most settings.

C NLP full per-setting breakdown

Table 4 reports the per-setting version of the main-text Tab. 1, restricted to the 7 method columns shared with the main-text breakdown (the pure TanGCE and TanGCE^{+} ablation rows of the main-text table are not repeated per-setting). For each of the 39 NLP settings and each of the four control-damage budgets, the signed deviation $(S - S_{\text{floor}})$ in pp at the trajectory step closest to floor. Bold marks the row-best per setting per budget block (every method within numeric precision of the row-best is bolded). The trailing rows summarize across settings: per-method mean and TanGCE^{++} W/T/L count vs. the best other method, with the 0.5pp tie band ensuring that any setting where the row-best is tied between TanGCE^{++} and another method counts as T (not W).

D CelebA full per-attribute breakdown

Tab. 5 is the full per-method $\times$ per-budget summary across both control regimes, of which the main-text Fig. 4 is the least-correlated $\Delta Y \in \{1, 3\}$pp slice. Cells: signed mean $(S - S_{\text{floor}})$ in pp at the trajectory step closest to chance; in-budget reach (denominator) and at-chance count (numerator) over the same.

Fig. 8 reports the per-attribute nonlinear-probe accuracy S for each of the 40 CelebA attributes at all four control-damage budgets, in the same bar-chart style as the NLP per-model figures (Fig. 3, App. Fig. 5–7): three bars per attribute (Clean / TanGCE^{++} / Obliviator) and a horizontal tick at the per-attribute majority-class floor. Hatched bars annotated "N/A" indicate that the method had no in-budget step at the shown ΔY budget; we substitute its clean (un-erased) S so the visual height communicates "failed to land" rather than a real measurement. A clear win for TanGCE^{++} reads as a low solid blue bar paired with a high (often hatched-N/A) orange Obliviator bar — common at $\Delta Y \leq 1$pp in the most-correlated regime, where Obliviator overshoots the budget on 39 of 40 entangled targets while TanGCE^{++} lands at floor on 15.

Tabs. 6–8 report the per-attribute numerical version of the main-text Tab. 5, one table per control-damage budget $\Delta Y \in \{1, 3, 5\}$pp: signed $(S - S_{\text{floor}})$ in pp at the trajectory step closest to the per-attribute majority-class floor, subject to mean $\Delta Y \geq -B$pp on the 5 controls. Bold marks the row-best per regime block. Trailing rows summarize per-regime per-method mean and TanGCE^{++} W/T/L count vs. the best other method (0.5pp tie band). The tightest budget ($\leq$ 1pp) is where the gap between TanGCE^{++} and the baselines is largest — it is also where Obliviator overshoots most often and shows "–" on the hardest entangled targets.

E Intuition: "fraction of the local spectrally-normalized neighborhood"

Section 5.4 introduces the per-sample trust region $\|\tilde{\mathbf{x}}_i - \mathbf{x}_i\|_2 \leq \varepsilon \cdot s_i$ and offers two choices for the local scale s_i: the simple radial k-NN distance r_i, or the spectrally-normalized tangent radius (Eq. 14). The paper describes the spectral choice as making ε "a fraction of the local spectrally-normalized neighborhood." This appendix unpacks what that phrase means in concrete terms.

Reading the trust region as a budget. With $s_i = r_i$ (the radial choice), the bound reads "move at most $\varepsilon\cdot$ Euclidean k-NN radius." This is direction-agnostic: every direction inside ambient space gets the same budget. With s_i defined by Eq. 14 (the spectrally-normalized tangent radius), the bound instead reads "move at most ε of the neighborhood radius *after dividing each local tangent axis by its own singular value* $\sigma_{i,k}$." That distinction is what makes ε direction-aware. The construction is Mahalanobis covariance normalization expressed through the local SVD: in the tangent basis, $\mathbf{C}_i^{-1/2}$ acts diagonally as $\sigma_{i,k}^{-1}$, so dividing each tangent coordinate by $\sigma_{i,k}$ is the same operation, written in the coordinate system the secant decomposition already gave us.

Worked example. Suppose the local tangent space at $\mathbf{x}_i$ is two-dimensional with right singular vectors $\mathbf{v}_1, \mathbf{v}_2$ and singular values $\sigma_{i,1} = 10$, $\sigma_{i,2} = 1$. Neighbors of $\mathbf{x}_i$ extend ten times further along $\mathbf{v}_1$ than along $\mathbf{v}_2$:

$\mathbf{v}_1$ direction: long / fat / well-supported, $\mathbf{v}_2$ direction: short / thin / fragile.

The whitened tangent coordinate of a neighbor offset $\mathbf{x}_j - \mathbf{x}_i$ along $\mathbf{v}_k$ is $z_k = \langle \mathbf{x}_j - \mathbf{x}_i, \mathbf{v}_k \rangle / \sigma_{i,k}$. Moving 1 ambient unit along $\mathbf{v}_1$ is therefore $1/10 = 0.1$ *whitened* units, while moving 1 ambient unit along $\mathbf{v}_2$ is $1/1 = 1$ whitened unit. Expressed in whitened coordinates the local neighborhood is approximately spherical: "10% of the whitened neighborhood" is the same allowed move *regardless of direction inside the tangent*. Translated back through the singular spectrum, that allowed move is large in ambient norm along $\mathbf{v}_1$ and small along $\mathbf{v}_2$, which is exactly the asymmetry we want: the linear approximation is reliable along high-σ axes and fragile along low-σ ones.

Comparison with the Euclidean radius. The Euclidean form is the inequality

$$\|\boldsymbol{\delta}\|_2 \;\leq\; \varepsilon \cdot r_i,$$

which treats every direction equally. The whitened form is approximately

$$\left\| \mathbf{B}_i^\top \boldsymbol{\delta} \,/\, \boldsymbol{\sigma}_i \right\|_2 \;\leq\; \varepsilon \cdot R_i^{\text{white}},$$

where division is component-wise and R_i^{spec} is the median neighbour displacement in spectrally-normalized tangent coordinates (Eq. 14). Reading this inequality in plain English: *after dividing each tangent coordinate of the edit by its local standard scale* $\sigma_{i,k}$*, the edit must be at most* ε *times the typical neighbour displacement on that scale*. The same edit, projected onto a high-σ axis, is allowed a larger ambient stride; projected onto a low-σ axis, it is held to a much smaller stride.

Mental model. Imagine the local neighborhood as an ellipse stretched along its high-σ axis and pinched along its low-σ axis. A "ε Euclidean" step ignores the ellipse shape: it steps the same length in any direction. A "ε spectrally-normalized" step instead asks: *after reshaping the ellipse into a sphere via* $\sigma_{i,k}^{-1}$ *scaling, does the edit fit inside an* ε *ball?* The radius is then mapped back into the original representation space. The result is a direction-aware trust region: it prevents the iterative loop from taking a step that is numerically small in Euclidean norm but huge relative to a thin local tangent direction where the linear approximation is the least reliable.

This is the geometric reason why $\varepsilon = 0.1$ is interpretable as a smaller knob than a single global step size λ would be: it bounds the edit relative to the local neighborhood's natural scale on each axis, not to a setting-independent constant.

Why ε transfers across datasets without per-setting tuning. A direct consequence of the whitening is that ε no longer carries the units of the local representation geometry: each tangent axis is rescaled by its own $\sigma_{i,k}$ before ε is applied, so a representation setting with a wide local spread along its dominant axes and one with a narrow local spread both report the same allowed move when expressed as a fraction of their respective whitened neighborhoods. This is why we can fix $\varepsilon = 0.1$ once and use it across all 39 NLP settings (13 language models $\times$ 3 concepts at 50%-depth) and all 80 CelebA settings (40 attributes $\times$ 2 control sets) without any per-setting tuning of the trust-region radius. Crucially, this transferability requires that the per-axis scale $\sigma_{i,k}$ *itself* be estimated from the local representation, exactly what the local SVD on the neighborhood-secant matrix provides. A single global step size cannot have this property because its units are setting-specific: the same constant moves a Gemma-3-27B representation by a much larger ambient distance than it does a Qwen2.5-0.5B one.

F CelebA control-set construction (least- vs. most-correlated five attributes)

For each of the 40 binary CelebA attributes used as a target concept, we report two complementary control regimes (Section 7.4, Tabs. 6–8, Figure 8): the *5 least-correlated* attributes and the *5 most-correlated* attributes, both selected from the remaining 39 attributes. This appendix records exactly how the two control sets are computed.

Computation. Let $\mathbf{Y} \in \{0,1\}^{N_{\text{train}} \times 40}$ be the train-split label matrix, with $\mathbf{Y}_{i,k}$ the label of training example i on attribute k. We compute the empirical Pearson correlation matrix

$$\mathbf{R}_{k\ell} \;=\; \frac{\mathrm{Cov}(\mathbf{Y}_{:,k}, \mathbf{Y}_{:,\ell})}{\sqrt{\mathrm{Var}(\mathbf{Y}_{:,k}) \cdot \mathrm{Var}(\mathbf{Y}_{:,\ell})}} \quad \text{for } k, \ell \in \{1, \ldots, 40\}. \tag{16}$$

For each target attribute k, we then rank the remaining 39 attributes by absolute correlation $|\mathbf{R}_{k\ell}|$:

- the *5 least-correlated* controls are the bottom 5 by $|\mathbf{R}_{k\ell}|$. They share *minimal* information with the target and isolate clean surgicality.
- the *5 most-correlated* controls are the top 5 by $|\mathbf{R}_{k\ell}|$. They share *substantial* information with the target and form a stricter test of whether erasure can preserve genuinely related downstream features.

The target attribute itself is excluded from the candidate pool (i.e., $\ell \neq k$). Pairs are computed in float64 with no shrinkage.

Concrete examples. For *Male*, the 5 most-correlated attributes are *Wearing_Lipstick* ($|r| \approx 0.78$), *No_Beard*, *Heavy_Makeup*, *Attractive*, and *5_o_Clock_Shadow*; the 5 least-correlated are *Narrow_Eyes*, *Blurry*, *Straight_Hair*, *Receding_Hairline*, and *Pale_Skin* (all $|r| < 0.05$). Erasing *Male* while preserving the most-correlated controls is genuinely hard because the controls are partial proxies for the target; erasing while preserving the least-correlated controls tests whether the intervention is geographically narrow but not whether it disentangles correlated information. We therefore report both regimes side by side: each one isolates a different failure mode.

Full per-target listing. Tabs. 9–10 list the two control sets for every one of the 40 target attributes, each with its absolute Pearson $|r|$ on the train split. They are reproduced from `outputs/celeba_correlated_controls.json`, the precomputed file consumed by every CelebA run in this paper.

Why two regimes per concept. Reporting only one control set hides whichever regime the method happens to be weak in. A method that destroys the entire representation will pass the least-correlated test (controls are essentially independent of the edit anyway) and fail the most-correlated test catastrophically; conversely, a method that performs no erasure will pass both. The difference

between the two regimes per method is therefore informative even when both individual numbers look acceptable. Section 7.4 reads the difference as a measure of how much an erasure method confuses "avoid this concept" with "avoid everything correlated with it".

G Closed-form per-row λ_i statistics

For each setting and each outer iteration of the proposed method (Phase 2 with fixed trust-region $\varepsilon = 0.1$), the closed-form rule of Eq. 13 produces a per-row step magnitude λ_i (capped at $\lambda_{\max} = 64$). Table 11 reports the distribution of these per-row λ_i averaged across all 60 iterations of each trajectory and then aggregated across settings.

The mean across the entire grid, $\bar{\lambda} \approx 29.31$, is the value at which AmbGCE^{++} appears in Tab. 1: forcing the unconstrained ambient method to take the same effective step magnitude that the closed-form per-row rule produces under TanGCE^{++} isolates the gradient-direction constraint (tangent-projected vs. ambient) as the only remaining axis of variation between the two methods.

H Anisotropy regimes for the local-σ spectrum

Eq. 15 reweights each tangent coordinate $a_{i,k}$ by $\sigma_{i,k}^{\alpha}$, where $\alpha \in \mathbb{R}$ is an anisotropy exponent and $\sigma_{i,k}$ are the singular values of the local secant matrix $\mathbf{S}_i$ (Eq. 7). Three non-negative values of α have natural interpretations:

- $\alpha = 0$: the update is isotropic in the tangent basis and reduces to Eq. 11. Reported as an ablation.
- $\alpha = 1$: each tangent direction is reweighted by $\sigma_{i,k}$. **This is the default in this paper.** Equivalently, the update applies the positive square root of the local secant covariance, $\mathbf{C}_i^{1/2}$, since the eigenvalues of $\mathbf{C}_i = \frac{1}{k}\mathbf{S}_i^\top \mathbf{S}_i$ are proportional to $\sigma_{i,k}^2$.
- $\alpha = 2$: each tangent direction is reweighted by $\sigma_{i,k}^2$, equivalent to applying $\mathbf{C}_i$. This is the natural-gradient / full-Mahalanobis trust-region optimum: it is the first-order solution of $\min_{\boldsymbol{\delta}} \langle \mathbf{g}_i, \boldsymbol{\delta} \rangle$ subject to $\boldsymbol{\delta}^\top \mathbf{C}_i^{-1} \boldsymbol{\delta} \leq \varepsilon^2$. Reported as an ablation; risks suppressing low-variance signal that may carry residual concept information.

Relation to Mahalanobis whitening (and what $\alpha = 1$ is *not*). A reader familiar with Mahalanobis whitening might wonder whether our $\alpha = 1$ is the same operation. It is not. Mahalanobis whitening transforms the data so the covariance becomes identity, $\mathbf{x}^{\text{white}} = \mathbf{C}^{-1/2}\mathbf{x}$, and corresponds to scaling each tangent direction by $\sigma_{i,k}^{-1}$, i.e. $\alpha = -1$ in our notation. Whitening makes every direction equal-weighted, which would dilute the concept-specific signal we want to suppress and force every tangent axis to receive equal step magnitude.

We use a *positive* power, $\sigma_{i,k}^{+1}$, which does the opposite: it reweights the tangent step *toward* high-σ principal axes (the directions where the local manifold has the most support and where the linear approximation is most reliable) and away from low-σ thin tail directions where the linear approximation is fragile and motion most easily leaves the manifold. The three non-negative values $\alpha \in \{0, 1, 2\}$ form a one-parameter family interpolating between (i) treating every tangent direction as equally trustworthy ($\alpha = 0$, isotropic), (ii) reweighting by $\sigma_{i,k}$ to respect the local ellipsoid mildly ($\alpha = 1$, our default), and (iii) the natural-gradient / Mahalanobis trust-region optimum that maximally suppresses thin directions ($\alpha = 2$). Whitening ($\alpha = -1$) sits outside this range for our problem because its sign is opposite the geometric correction we want; we therefore do not include it.

I Coordinate-chart definition of the tangent space

We adopt the formal definition from Tu [2011]. Fix $\mathbf{x} \in \mathcal{M}$. A small local neighborhood $U \subset \mathcal{M}$ around $\mathbf{x}$ admits a coordinate map $\varphi : U \to \Omega \subset \mathbb{R}^m$ with $\varphi(\mathbf{x}) = 0$. The inverse $\varphi^{-1} : \Omega \to U \subset \mathbb{R}^d$ is a local parametrization back into the ambient hidden space. The tangent space at $\mathbf{x}$ is the m-dimensional linear subspace

$$T_{\mathbf{x}}\mathcal{M} = d(\varphi^{-1})_0(\mathbb{R}^m) \subset \mathbb{R}^d, \tag{17}$$

where $d(\varphi^{-1})_0$ is the differential of φ^{-1} at the origin (its Jacobian, in ML language). It maps an infinitesimal coordinate displacement in $\mathbb{R}^m$ to the corresponding first-order ambient displacement. Because $\mathcal{M}$ is embedded in $\mathbb{R}^d$, $T_{\mathbf{x}}\mathcal{M}$ is an m-dimensional linear subspace of the ambient hidden space.

In TanGCE we never construct φ or φ^{-1}. The leading right singular vectors of the local secant matrix (Eq. 7) serve as an empirical first-order basis for $T_{\mathbf{x}}\mathcal{M}$, and the orthogonal projector $\mathbf{P}_{T_i} = \mathbf{B}_i\mathbf{B}_i^\top$ implements the tangent component $\mathbf{g}^\top$ of any ambient vector $\mathbf{g}$. We do not claim that $\mathcal{M}$ is smooth in any theorem-level sense, only that this approximation tracks the local sample structure well enough to attenuate updates in directions unsupported by it.

Table 3: **Phase-1 closed-form ablations on the 39-panel NLP grid (50%-depth layer).** Same fresh nonlinear probe ($h = 128$, 200 SGD steps, patience 3) is trained on each post-erasure representation. *LEACE* is the affine LEACE eraser. *LEACE + CovMatch* additionally projects out the top-2 eigenvectors of $\Delta\Sigma = \Sigma_+ - \Sigma_-$ (rank-3 closed-form erasure total). *TanGCE*$^{++}$ is the proposed full pipeline (LEACE + CovMatch preprocessing + iterative tangent loop, ε=0.1); we report the best $|S-\text{floor}|$ trajectory step subject to $\Delta Y \geq -5$pp. NL = nonlinear probe accuracy on test (lower is better; floor is the per-panel majority class). ΔY in pp on the control task. Bold marks the row's NL minimum.

Model	Concept	Floor (NL*)	LEACE NL	LEACE ΔYpp	LEACE + CovMatch NL	LEACE + CovMatch ΔYpp	TanGCE^{++} ε=0.1 NL	TanGCE^{++} ε=0.1 ΔYpp
Qwen2.5-0.5B	Sycophancy	0.550	**0.550**	+0.0	**0.550**	-2.1	**0.550**	-0.1
	Gender	0.539	0.709	-0.1	0.684	-0.4	**0.549**	-4.1
	Safety	0.510	0.752	+0.0	0.529	+0.0	**0.510**	-0.1
Qwen2.5-1.5B	Sycophancy	0.550	**0.550**	-0.1	**0.550**	-1.5	**0.550**	+1.2
	Gender	0.539	0.822	-0.3	0.788	-1.7	**0.592**	-3.9
	Safety	0.510	0.759	+2.9	**0.510**	+2.9	**0.510**	+0.0
Qwen2.5-3B	Sycophancy	0.675	**0.675**	+0.1	**0.675**	-2.7	**0.675**	-0.7
	Gender	0.539	0.857	-1.3	0.783	-0.1	**0.548**	-4.1
	Safety	0.510	0.718	+0.5	0.520	+0.5	**0.510**	+0.0
Gemma-2-2B	Sycophancy	0.543	0.643	+0.0	**0.543**	-0.4	**0.543**	-0.2
	Gender	0.539	0.722	+0.3	0.681	-0.7	**0.539**	-4.9
	Safety	0.510	0.715	+0.7	0.654	+0.7	**0.510**	+0.0
Gemma-2-9B	Sycophancy	0.569	0.751	-0.1	**0.569**	-0.7	0.569	+0.0
	Gender	0.539	0.638	+0.2	0.562	-0.4	**0.539**	-3.5
	Safety	0.510	0.623	+0.0	**0.510**	+0.0	**0.510**	+0.0
Gemma-2-27B	Sycophancy	0.553	0.695	-0.2	**0.553**	-0.9	**0.553**	-0.3
	Gender	0.539	0.663	+0.0	0.605	-2.5	**0.539**	-0.5
	Safety	0.510	0.736	+0.7	0.643	+1.4	**0.510**	+0.0
Llama-3.2-1B	Sycophancy	0.551	0.849	-0.1	**0.551**	-0.2	**0.551**	+0.0
	Gender	0.539	0.855	-0.4	0.738	+0.8	**0.536**	-3.9
	Safety	0.510	0.817	+0.3	0.778	+0.3	**0.498**	+0.0
Llama-3.2-3B	Sycophancy	0.551	0.896	-0.0	**0.551**	+0.0	**0.551**	+0.0
	Gender	0.539	0.889	+0.2	0.794	-0.4	**0.535**	-2.8
	Safety	0.510	0.837	+0.1	0.821	+0.2	**0.526**	+0.0
Mistral-7B	Sycophancy	0.551	0.920	+0.0	**0.551**	-0.1	**0.551**	+0.2
	Gender	0.539	0.851	-0.4	0.815	-0.3	**0.606**	-4.8
	Safety	0.510	0.825	+0.0	0.811	+0.0	**0.532**	+0.2
Gemma-3-1B	Sycophancy	0.551	0.727	-0.1	**0.551**	-2.2	**0.551**	-0.0
	Gender	0.539	0.708	+0.0	0.613	-0.8	**0.539**	+2.0
	Safety	0.510	0.726	+0.0	0.723	-1.0	**0.510**	+1.0
Gemma-3-4B	Sycophancy	0.551	**0.551**	-0.0	**0.551**	-0.9	**0.551**	-0.4
	Gender	0.539	0.704	-0.5	0.580	-2.7	**0.539**	+4.9
	Safety	0.510	0.735	+1.0	0.609	+1.0	**0.510**	-0.2
Gemma-3-12B	Sycophancy	0.556	0.613	+0.0	**0.556**	-0.5	**0.556**	-0.5
	Gender	0.539	**0.539**	+3.5	0.606	-8.7	0.539	+3.1
	Safety	0.510	0.589	+3.8	**0.510**	+3.8	**0.510**	+0.0
Gemma-3-27B	Sycophancy	0.526	0.586	-0.2	0.566	-0.1	**0.526**	+0.0
	Gender	0.539	0.700	+1.7	0.678	-2.6	**0.539**	-4.8
	Safety	0.510	0.686	+0.9	**0.510**	+0.7	**0.510**	+0.0
Mean sycophancy		0.560	0.693	-0.0	0.563	-0.9	**0.560**	-0.1
Mean gender		0.539	0.743	+0.2	0.687	-1.6	**0.549**	-2.1
Mean safety		0.510	0.732	+0.8	0.625	+0.8	**0.512**	+0.1

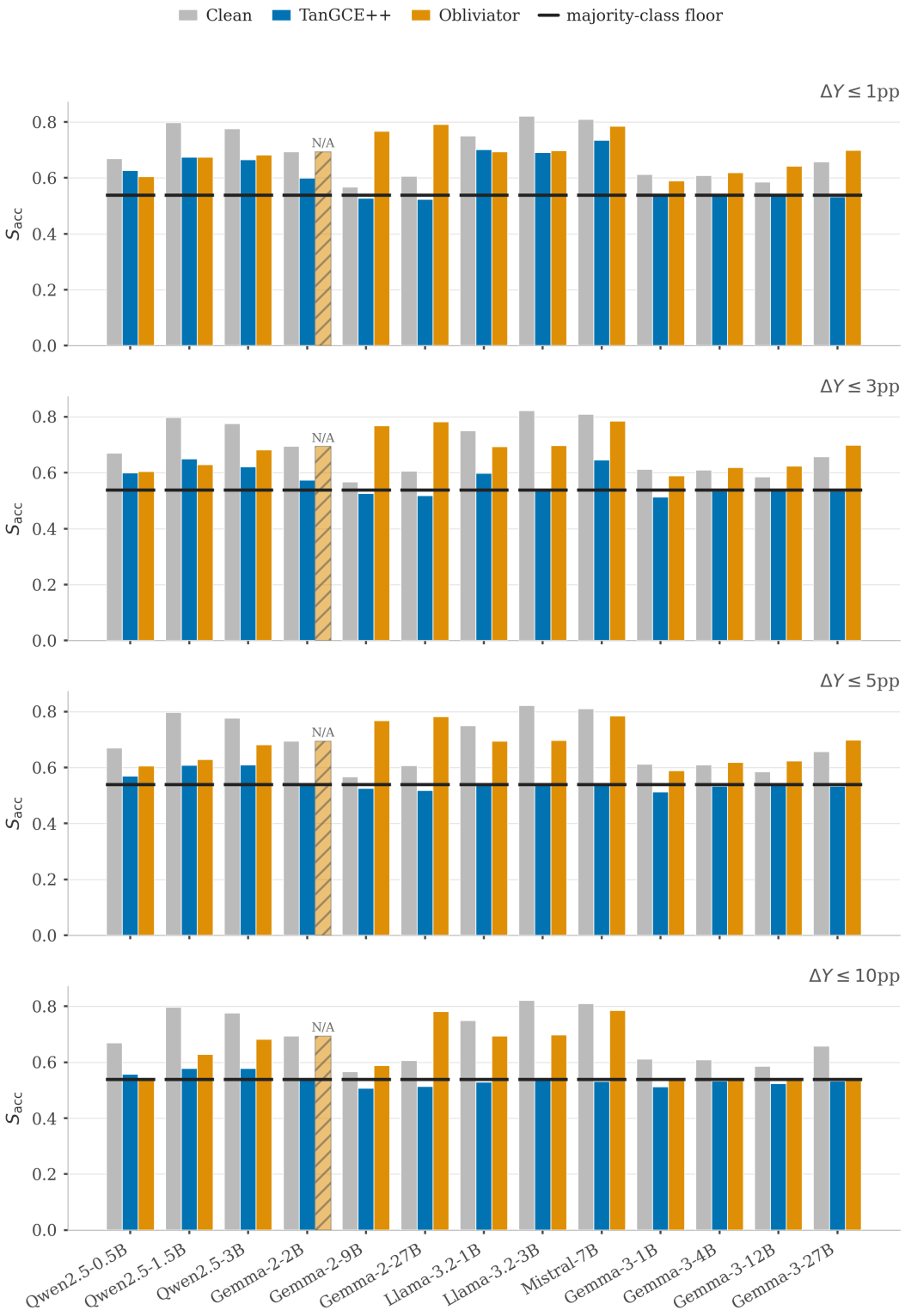


Figure 5: **Gender (full): per-model nonlinear-probe accuracy at all four control-damage budgets.** Full four-budget version of the slim two-budget figure in the main text (Fig. 3). Each row is one budget $\Delta Y \in \{1, 3, 5, 10\}$pp.

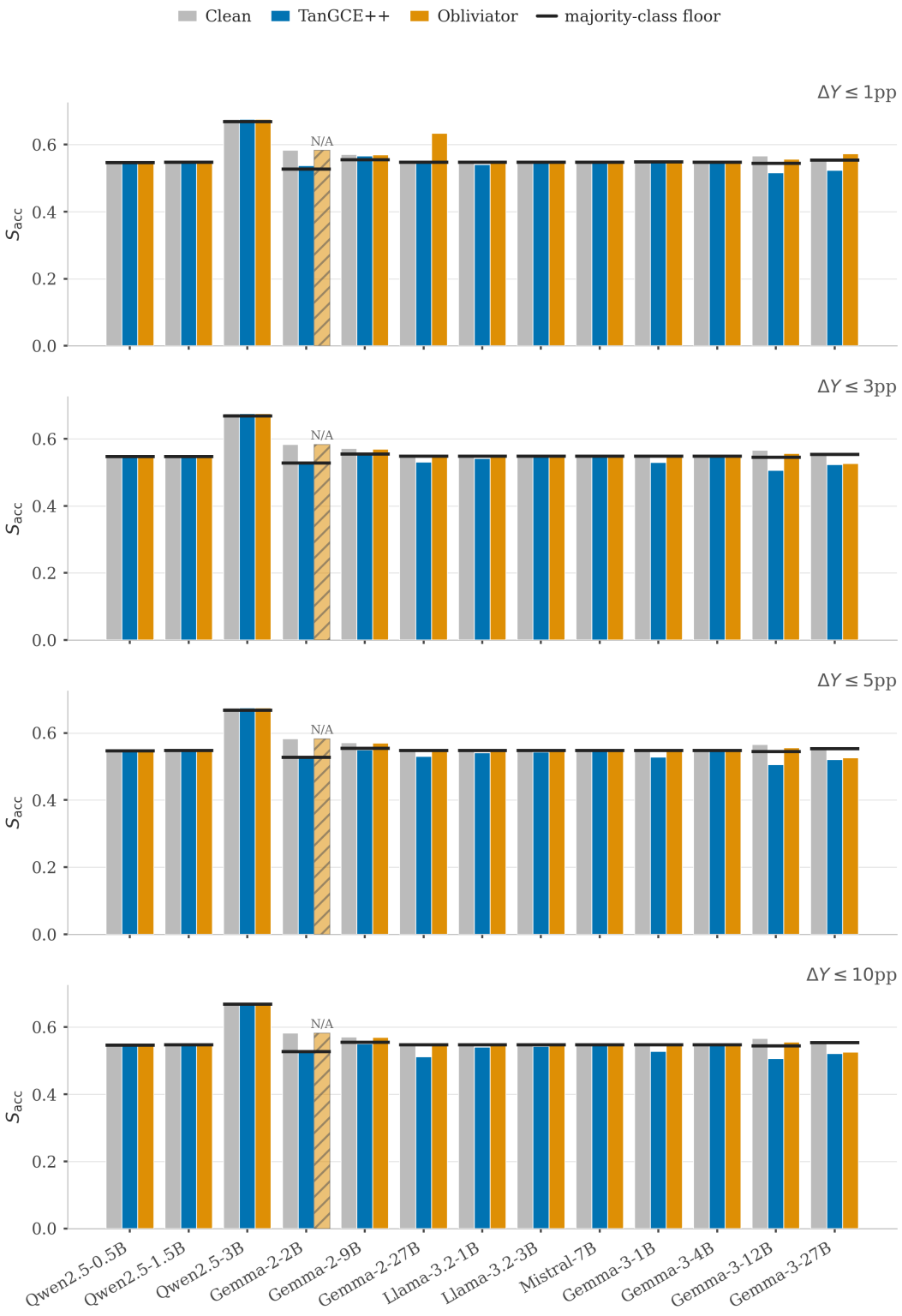


Figure 6: **Sycophancy: per-model nonlinear-probe accuracy at four control-damage budgets.** Layout matches Fig. 5.

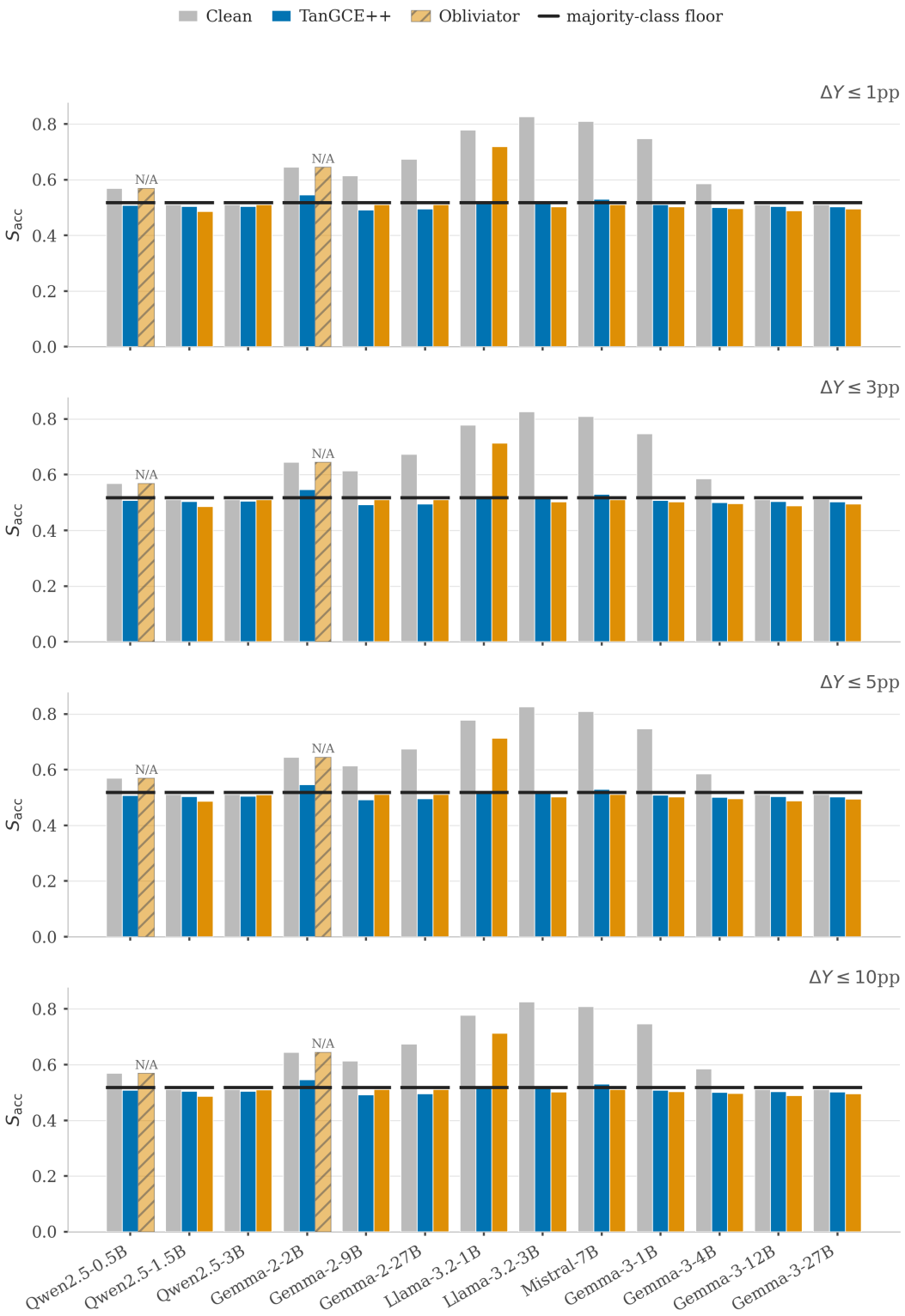


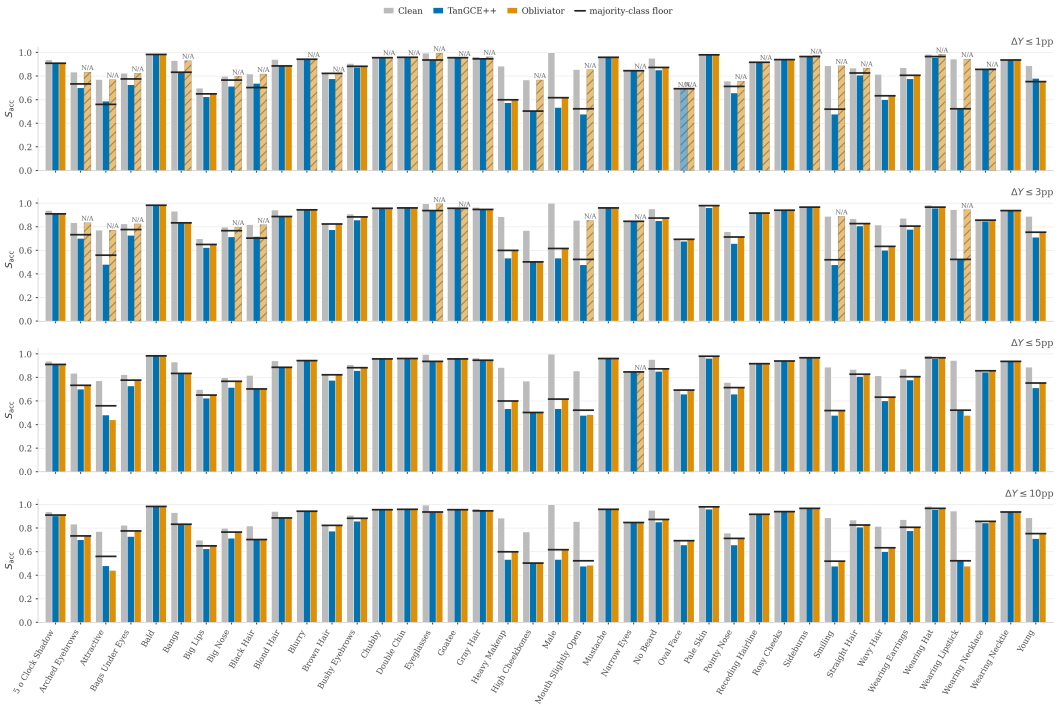
Figure 7: **Safety: per-model nonlinear-probe accuracy at four control-damage budgets.** Layout matches Fig. 5.

Table 4: **Per-panel breakdown.** Signed $(S - S_{\text{floor}})$ in pp at the trajectory step closest to the per-panel majority-class floor, subject to $|\Delta Y| \leq B$pp control-damage budget. Bold = best (smallest absolute deviation from floor) per panel per budget block. '–' = no in-budget step at that budget for that method. Final rows: per-budget per-method mean across the 39 panels and TanGCE^{++} W/T/L counts vs. the best other method (0.5pp tie band).

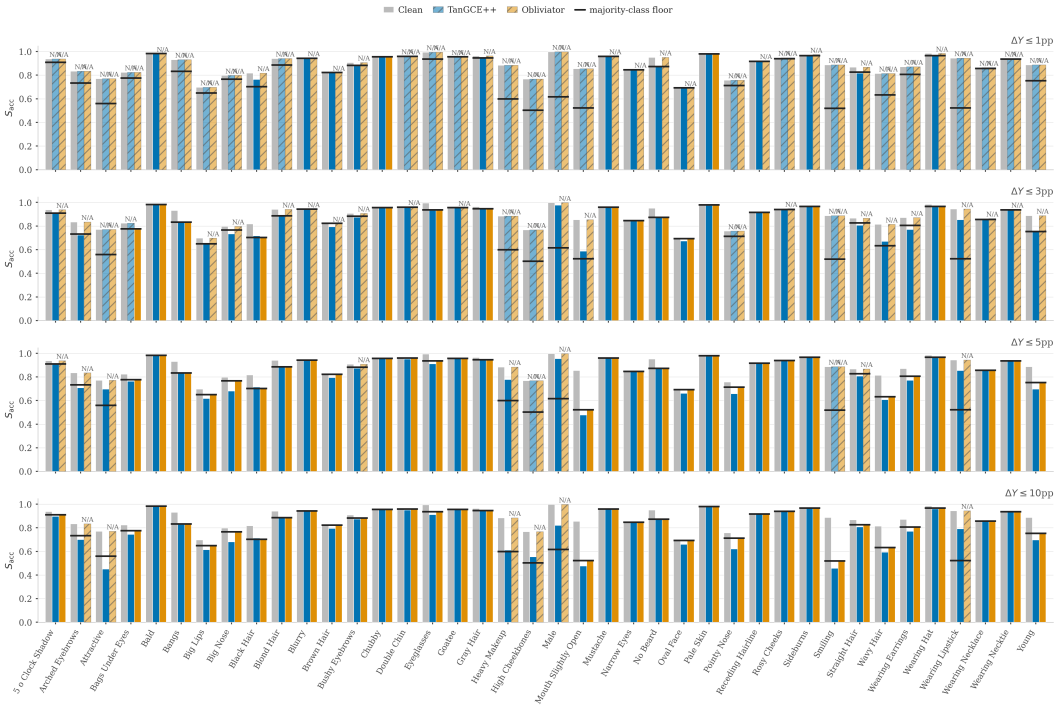
Panel	$\Delta Y \leq$ 1pp							$\Delta Y \leq$ 3pp							$\Delta Y \leq$ 5pp							$\Delta Y \leq$ 10pp						
	LE	L+C	INLP	IGBP	Obl	Amb++	Tan++	LE	L+C	INLP	IGBP	Obl	Amb++	Tan++	LE	L+C	INLP	IGBP	Obl	Amb++	Tan++	LE	L+C	INLP	IGBP	Obl	Amb++	Tan++
Gemma2-27B/Gen L23	+12.3	–	+11.1	+12.8	**+0.0**	–	**+0.0**	+12.3	+6.6	+11.1	+12.8	**+0.0**	–	**+0.0**	+12.3	+6.6	+11.1	+12.8	**+0.0**	–	**+0.0**	+12.3	+6.6	+11.1	+12.8	**+0.0**	–	**+0.0**
Gemma2-27B/Saf L23	+22.5	+13.3	+28.6	+22.3	-0.7	-1.7	**-0.1**	+22.5	+13.3	+28.6	+22.3	-0.7	-1.0	**-0.1**	+22.5	+13.3	+28.6	+22.3	-0.7	**-0.1**	-0.1	+22.5	+13.3	+28.6	+22.3	-0.7	**-0.1**	-0.1
Gemma2-27B/Syc L23	+14.3	**+0.0**	+10.7	+14.8	+8.5	+18.7	+0.5	+14.3	**+0.0**	+10.7	+14.8	+0.5	+18.7	+0.5	+14.3	+0.0	+10.7	+14.8	+0.5	+15.5	**-0.0**	+14.3	+0.0	+10.7	+14.8	+0.5	+15.5	**-0.0**
Gemma2-2B/Gen L13	+18.3	+14.2	+15.1	**+5.6**	+24.0	+18.5	+6.0	+18.3	+14.2	+15.1	+5.6	+21.8	+18.5	**+3.4**	+18.3	+14.2	+15.1	+5.6	+21.8	+18.5	**+0.0**	+18.3	+14.2	+15.1	+5.6	+21.8	+18.5	**+0.0**
Gemma2-2B/Saf L13	+20.5	+14.3	+12.6	+6.7	-0.6	**+0.1**	+2.8	+20.5	+14.3	+12.6	+6.7	-0.6	**+0.1**	+2.8	+20.5	+14.3	+12.6	+6.7	-0.6	**+0.1**	+2.8	+20.5	+14.3	+12.6	+6.7	-0.6	**-0.0**	+2.8
Gemma2-2B/Syc L13	+9.9	**+0.0**	+2.5	+2.8	+1.6	+13.0	+0.9	+9.9	**+0.0**	+2.5	+2.8	+1.6	+13.0	+0.2	+9.9	**+0.0**	+2.5	+2.8	+1.6	+13.0	+0.2	+9.9	**+0.0**	+2.5	+2.8	+1.6	+12.9	+0.2
Gemma2-9B/Gen L21	+9.9	+2.3	+5.1	+18.6	+17.4	+15.4	**+0.0**	+9.9	+2.3	+5.1	+13.7	+17.4	+15.4	**+0.0**	+9.9	+2.3	+5.1	+13.7	+17.4	+15.4	**+0.0**	+9.9	+2.3	+5.1	+13.7	+4.8	+15.4	**+0.0**
Gemma2-9B/Saf L21	+11.3	**+0.0**	+16.5	+7.2	-0.4	+2.0	+0.0	+11.3	**+0.0**	+16.5	+7.2	-0.4	+0.1	+0.0	+11.3	**+0.0**	+16.5	+7.2	-0.4	+0.1	+0.0	+11.3	**+0.0**	+16.5	+7.2	-0.4	+0.1	+0.0
Gemma2-9B/Syc L21	+18.2	**-0.0**	+7.5	+10.6	+1.4	+22.1	+1.0	+18.2	**-0.0**	+7.5	+10.6	+1.4	+22.1	+0.2	+18.2	**-0.0**	+7.5	+10.6	+1.4	+22.1	+0.2	+18.2	**-0.0**	+7.5	+10.6	+1.4	+20.9	+0.2
Gemma3-12B/Gen L24	-0.0	–	–	–	**+0.0**	+8.8	**+0.0**	-0.0	–	+11.0	–	**+0.0**	-0.9	**+0.0**	-0.0	–	+10.3	–	**+0.0**	**+0.0**	**+0.0**	-0.0	+6.7	+6.3	–	**+0.0**	**+0.0**	**+0.0**
Gemma3-12B/Saf L24	+7.9	**+0.0**	+15.4	+24.5	-0.7	+0.1	+0.1	+7.9	**+0.0**	+15.4	+24.5	-0.7	+0.1	+0.1	+7.9	**+0.0**	+15.4	+24.5	-0.7	+0.1	+0.1	+7.9	**+0.0**	+15.4	+24.5	-0.7	+0.1	+0.1
Gemma3-12B/Syc L24	+5.7	**+0.0**	+0.8	+4.4	+1.1	–	+0.4	+5.7	**+0.0**	+0.8	+4.4	+1.1	+7.7	+0.1	+5.7	**+0.0**	+0.8	+4.4	+1.1	+7.7	+0.0	+5.7	**+0.0**	+0.8	+4.4	+1.1	+7.7	+0.0
Gemma3-1B/Gen L13	+16.9	+7.4	+8.5	–	+5.0	–	**+0.0**	+16.9	+7.4	+2.1	+15.0	+5.0	+15.4	**+0.0**	+16.9	+7.4	**+0.0**	+12.7	+5.0	+15.4	**+0.0**	+16.9	+7.4	**+0.0**	+12.7	**+0.0**	+15.4	**+0.0**
Gemma3-1B/Saf L13	+21.5	+21.2	+16.2	+20.6	-0.8	-0.4	**+0.1**	+21.5	+21.2	+16.2	+20.6	-0.8	-0.4	**+0.1**	+21.5	+21.2	+16.2	+20.6	-0.8	-0.4	**+0.1**	+21.5	+21.2	+16.2	+20.6	-0.8	-0.3	**+0.1**
Gemma3-1B/Syc L13	+17.6	–	+1.0	+4.9	+0.3	+0.3	**+0.2**	+17.6	**+0.0**	+1.0	+4.9	+0.3	+0.3	-0.1	+17.6	**+0.0**	+1.0	+4.9	+0.3	+0.3	-0.1	+17.6	**+0.0**	+1.0	+4.9	+0.3	+0.3	-0.1
Gemma3-27B/Gen L31	+16.0	–	+25.6	+23.9	+15.9	–	**+0.0**	+16.0	+13.9	+25.6	+23.9	+15.9	–	**+0.0**	+16.0	+13.9	+25.6	+23.9	+15.9	–	**+0.0**	+16.0	+13.9	+24.4	+23.9	**+0.0**	–	**+0.0**
Gemma3-27B/Saf L31	+17.6	**+0.0**	+23.8	+25.3	-0.3	-0.0	-0.2	+17.6	**+0.0**	+23.8	+25.3	-0.3	-0.0	-0.2	+17.6	**+0.0**	+23.8	+25.3	-0.3	-0.0	-0.2	+17.6	**+0.0**	+23.8	+25.3	-0.3	-0.0	-0.2
Gemma3-27B/Syc L31	+6.0	+4.0	**+0.3**	-0.3	+1.8	+4.3	**+0.3**	+6.0	+4.0	+0.3	-0.3	+1.8	+4.3	**+0.2**	+6.0	+4.0	+0.3	-0.3	+1.8	+4.3	**+0.2**	+6.0	+4.0	+0.3	-0.3	+1.8	+3.5	**+0.2**
Gemma3-4B/Gen L17	+16.5	–	+8.5	+19.5	+7.9	–	**+0.0**	+16.5	+4.0	+8.5	+19.5	+7.9	+15.0	**+0.0**	+16.5	+4.0	+8.5	+19.5	+7.9	+15.0	**+0.0**	+16.5	+4.0	+8.5	+19.5	**+0.0**	+15.0	**+0.0**
Gemma3-4B/Saf L17	+22.5	+9.9	+27.3	+26.1	-0.8	-0.1	**-0.0**	+22.5	+9.9	+27.3	+26.1	-0.8	-0.1	**-0.0**	+22.5	+9.9	+27.3	+26.1	-0.8	-0.1	**-0.0**	+22.5	+9.9	+27.3	+26.1	-0.8	-0.1	**-0.0**
Gemma3-4B/Syc L17	**+0.0**	**+0.0**	+0.3	+13.2	+0.2	–	+0.3	**+0.0**	**+0.0**	+0.3	+13.2	+0.2	+0.4	+0.3	**+0.0**	**+0.0**	+0.3	+13.2	+0.2	+0.4	-0.0	**+0.0**	**+0.0**	+0.3	+13.2	+0.2	+0.4	-0.0
Llama3.2-1B/Gen L8	+31.6	+19.9	+21.5	**+4.6**	+15.4	–	+16.2	+31.6	+19.9	+21.5	**+2.3**	+15.4	–	+5.8	+31.6	+19.9	+21.5	+2.3	+15.4	+25.7	**+0.0**	+31.6	+19.9	+21.5	+2.3	+15.4	+25.7	**+0.0**
Llama3.2-1B/Saf L8	+30.7	+26.7	+28.3	+8.8	-0.7	-5.7	**+0.1**	+30.7	+26.7	+28.3	+8.8	-0.7	-5.7	**+0.1**	+30.7	+26.7	+28.3	+8.8	-0.7	-5.7	**+0.1**	+30.7	+26.7	+28.3	+8.8	-0.7	-1.7	**+0.1**
Llama3.2-1B/Syc L8	+29.8	**+0.0**	+0.3	+0.3	+0.3	–	-0.0	+29.8	**+0.0**	+0.3	+0.3	+0.3	–	-0.0	+29.8	**+0.0**	+0.3	+0.3	+0.3	–	-0.0	+29.8	**+0.0**	+0.3	+0.3	+0.3	–	-0.0
Llama3.2-3B/Gen L14	+35.0	+25.5	+31.7	**+2.5**	+15.8	–	+15.1	+35.0	+25.5	+31.7	+2.1	+15.8	+30.3	**-0.1**	+35.0	+25.5	+31.7	+2.1	+15.8	+30.3	**-0.1**	+35.0	+25.5	+31.7	+2.1	+15.8	+30.3	**+0.0**
Llama3.2-3B/Saf L14	+32.7	+31.0	+33.1	+9.4	-0.8	-0.7	**-0.1**	+32.7	+31.0	+32.1	+9.4	-0.8	-0.7	**-0.1**	+32.7	+31.0	+32.1	+9.4	-0.8	-0.7	**-0.1**	+32.7	+31.0	+32.1	+9.4	-0.8	-0.7	**-0.1**
Llama3.2-3B/Syc L14	+34.5	**+0.0**	+0.3	+2.8	+0.3	–	+0.3	+34.5	**+0.0**	+0.3	+2.8	+0.3	–	+0.3	+34.5	**+0.0**	+0.3	+2.8	+0.3	–	+0.3	+34.5	**+0.0**	+0.3	+2.8	+0.3	+0.3	+0.3
Mistral-7B/Gen L16	+31.2	+27.6	+29.5	**+8.1**	+24.6	–	+19.5	+31.2	+27.6	+29.5	**+8.1**	+24.6	+28.6	+10.6	+31.2	+27.6	+29.5	+8.1	+24.6	+28.6	**+0.3**	+31.2	+27.6	+29.5	+8.1	+24.6	+28.6	**+0.0**
Mistral-7B/Saf L16	+31.5	+30.0	+34.6	+8.0	**-0.8**	-2.9	+1.1	+31.5	+30.0	+34.3	+8.0	**-0.8**	-2.9	+1.1	+31.5	+30.0	+34.3	+8.0	**-0.8**	-2.9	+1.1	+31.5	+30.0	+34.3	+8.0	**-0.8**	-2.9	+1.1
Mistral-7B/Syc L16	+37.0	**+0.0**	+0.3	+0.3	+0.3	+36.9	+0.3	+37.0	**+0.0**	+0.3	+0.3	+0.3	+36.7	+0.3	+37.0	**+0.0**	+0.3	+0.3	+0.3	+36.7	+0.3	+37.0	**+0.0**	+0.3	+0.3	+0.3	+36.7	+0.3
Qwen2.5-0.5B/Gen L12	+17.0	+14.5	+13.1	**+5.4**	+6.6	+15.8	+8.8	+17.0	+14.5	+13.1	**+5.4**	+6.6	+15.8	+5.9	+17.0	+14.5	+13.1	+5.4	+6.6	+15.8	**+3.0**	+17.0	+14.5	+13.1	+5.4	**+0.0**	+15.8	+1.8
Qwen2.5-0.5B/Saf L12	+24.1	+1.9	+16.7	+3.2	-0.8	+0.1	**+0.0**	+24.1	+1.9	+16.7	+3.2	-0.8	+0.1	**+0.0**	+24.1	+1.9	+16.7	+3.2	-0.8	+0.1	**+0.0**	+24.1	+1.9	+16.7	+3.2	-0.8	+0.1	**+0.0**
Qwen2.5-0.5B/Syc L12	**+0.0**	–	+0.3	+3.3	+0.3	–	+0.2	**+0.0**	**+0.0**	+0.3	+3.3	+0.3	–	+0.2	**+0.0**	**+0.0**	+0.3	+3.3	+0.3	–	+0.2	**+0.0**	**+0.0**	+0.3	+3.3	+0.3	–	+0.2
Qwen2.5-1.5B/Gen L14	+28.2	–	+22.3	**+4.1**	+13.5	+28.2	+13.5	+28.2	+24.9	+22.3	**+4.1**	+8.9	+28.2	+11.1	+28.2	+24.9	+22.3	**+4.1**	+8.9	+28.2	+6.8	+28.2	+24.9	+22.3	+4.1	+8.9	+28.2	**+3.9**
Qwen2.5-1.5B/Saf L14	+24.9	**+0.0**	+17.9	+6.0	-0.3	+2.7	-0.0	+24.9	**+0.0**	+17.9	+6.0	-0.3	+2.0	-0.0	+24.9	**+0.0**	+17.9	+6.0	-0.3	+2.0	-0.0	+24.9	**+0.0**	+17.9	+6.0	-0.3	+2.0	-0.0
Qwen2.5-1.5B/Syc L14	**+0.0**	–	-0.0	-0.4	+0.3	–	+0.3	**+0.0**	**+0.0**	-0.0	-0.4	+0.3	–	+0.3	**+0.0**	**+0.0**	-0.0	-0.4	+0.3	–	+0.3	**+0.0**	**+0.0**	-0.0	-0.4	+0.3	–	+0.3
Qwen2.5-3B/Gen L18	–	+24.4	+24.8	**+4.7**	+14.3	–	+12.6	+31.8	+24.4	+24.8	**+4.7**	+14.3	+26.5	+8.1	+31.8	+24.4	+24.8	**+4.7**	+14.3	+26.5	+7.1	+31.8	+24.4	+24.8	+4.7	+14.3	+26.5	**+3.8**
Qwen2.5-3B/Saf L18	+20.8	+0.9	+22.3	+15.4	-0.3	-0.2	**+0.0**	+20.8	+0.9	+22.3	+15.4	-0.3	**+0.0**	**+0.0**	+20.8	+0.9	+22.3	+15.4	-0.3	**+0.0**	**+0.0**	+20.8	+0.9	+22.3	+15.4	-0.3	**+0.0**	**+0.0**
Qwen2.5-3B/Syc L18	**+0.0**	–	+0.6	+0.6	+0.6	–	+0.6	**+0.0**	**+0.0**	+0.6	+0.6	+0.6	–	+0.6	**+0.0**	**+0.0**	+0.6	+0.6	+0.6	–	+0.6	**+0.0**	**+0.0**	+0.6	+0.6	+0.6	–	+0.6
Mean (pp)	+18.3	+9.6	+14.1	+9.5	+4.3	+7.3	+2.6	+18.6	+8.9	+13.8	+9.4	+4.0	+9.3	+1.3	+18.6	+8.9	+13.7	+9.4	+4.0	+9.7	+0.6	+18.6	+8.9	+13.6	+9.4	+2.7	+9.5	+0.4
TanGCE^{++} W/T/L				7/22/10							9/23/7							10/25/4							8/28/3			

Table 5: **CelebA cross-modal results (CLIP ViT-B/32, 40 binary attributes).** *Leakage (pp)*: signed $S - S_{\text{floor}}$ at the trajectory step closest to the per-attribute majority-class chance; 0 = at chance, lower in absolute value = stronger erasure. @*chance*: settings driven to chance ($|S - S_{\text{floor}}| \leq 0.5$pp) over settings the method reached the budget on. Each method is run on 40 attributes under each control regime (5 least- vs. 5 most-correlated controls per target). The CelebA pipeline uses the same TanGCE^{++} recipe as the NLP headline: LEACE preprocessing + rank-2 CovMatch + manifold loop. Best per column in **bold**. '–' = no in-budget step at that budget for that method.

Method	$\Delta Y \leq$ 1pp		$\Delta Y \leq$ 3pp		$\Delta Y \leq$ 5pp		$\Delta Y \leq$ 10pp	
	Leakage	@chance↑	Leakage	@chance	Leakage	@chance	Leakage	@chance
5 least-correlated controls (40 attributes).								
LEACE	+6.6	15/39	+6.4	16/40	+6.4	16/40	+6.4	16/40
LEACE+CovMatch	+5.4	21/37	+5.1	22/40	+5.1	22/40	+5.1	22/40
INLP	+5.5	26/40	+5.5	26/40	+5.5	26/40	+5.5	26/40
IGBP	+5.8	10/36	+5.3	13/40	+5.3	13/40	+5.3	13/40
Obliviator	**+0.0**	15/15	**+0.0**	29/29	−0.4	37/39	−0.4	38/40
AmbGCE^{++}	+1.4	15/20	+2.9	28/39	+0.7	32/40	+0.4	36/40
TanGCE^{++}	+0.2	35/39	**+0.0**	39/40	**+0.0**	39/40	**+0.0**	39/40
5 most-correlated controls (40 attributes).								
LEACE	+3.9	11/20	+6.2	16/37	+6.4	16/40	+6.4	16/40
LEACE+CovMatch	+0.5	9/12	+4.0	21/34	+4.7	22/39	+5.1	22/40
INLP	+5.8	26/40	+5.6	26/40	+5.5	26/40	+5.5	26/40
IGBP	+7.0	13/40	+5.6	13/40	+5.3	13/40	+5.3	13/40
Obliviator	**+0.0**	2/2	**+0.0**	15/15	**+0.0**	30/30	**+0.0**	34/34
AmbGCE^{++}	+2.2	1/3	+7.4	10/20	+5.8	17/33	+4.7	23/40
TanGCE^{++}	+0.4	15/16	+2.4	29/34	+2.6	33/38	+1.4	35/40



(a) 5 least-correlated controls (low entanglement).



(b) 5 most-correlated controls (high entanglement).

Figure 8: **CelebA per-attribute nonlinear-probe accuracy at four control-damage budgets.** Same bar-chart layout as the NLP per-model figures. For each attribute, three bars: Clean (gray, no erasure), TanGCE^{++} (blue, the proposed pipeline), Obliviator (orange, baseline). Black tick: per-attribute majority-class floor (lower bars are better; bars at the floor mean the probe is at chance). Hatched bars marked "N/A" = the method had no in-budget step at that ΔY; we substitute its clean (un-erased) S as a stand-in, so a hatched-orange tall bar next to a low solid-blue bar is a clean TanGCE^{++} win on a setting Obliviator could not reach. Most-correlated regime at $\Delta Y \leq 1$pp: $39/40$ Obliviator bars are hatched while TanGCE^{++} reaches the budget on $16/40$.

Table 6: **CelebA per-attribute breakdown at $\Delta Y \leq 1$pp (40 attributes $\times$ 2 control regimes).** Signed $(S - S_{\text{floor}})$ in pp at the trajectory step closest to the per-attribute majority-class floor, subject to mean $\Delta Y \geq -1$pp on the 5 controls. Bold = best (smallest absolute deviation from floor) per attribute per regime block. '–' = no in-budget step at that budget for that method. Final rows: per-regime per-method mean and TanGCE^{++} W/T/L counts vs. the best other method (0.5pp tie band).

Concept	5 least-correlated controls							5 most-correlated controls						
	LE	L+C	INLP	IGBP	Obl	Amb++	Tan++	LE	L+C	INLP	IGBP	Obl	Amb++	Tan++
5_o_Clock_Shadow	+3.0	+1.3	**0.0**	+3.0	**0.0**	+1.3	**0.0**	–	–	**0.0**	+3.0	–	–	–
Arched_Eyebrows	+9.7	+3.0	+6.7	+6.7	–	+4.7	**0.0**	–	–	**+6.7**	**+6.7**	–	–	–
Attractive	+15.3	+17.3	+17.7	**+1.0**	–	–	+2.7	–	–	**+17.0**	+18.0	–	–	–
Bags_Under_Eyes	+0.3	**0.0**	**0.0**	+0.3	–	–	**0.0**	–	–	**0.0**	+0.3	–	–	–
Bald	**0.0**	**0.0**	**0.0**	–	**0.0**	–	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	–	–	**0.0**
Bangs	+4.7	**0.0**	**0.0**	+3.7	–	–	**0.0**	–	–	**0.0**	+3.7	–	–	–
Big_Lips	+1.7	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	–	–	**0.0**	**0.0**	–	–	–
Big_Nose	+2.3	+2.3	**0.0**	–	–	–	**0.0**	–	–	**0.0**	+2.0	–	–	–
Black_Hair	+10.7	–	+8.3	+4.0	–	–	**+3.3**	+10.7	+7.7	+8.3	**+4.0**	–	–	+6.0
Blond_Hair	+5.0	+3.0	+3.0	+3.3	**0.0**	**0.0**	**0.0**	+5.0	–	**+3.0**	+3.3	–	–	–
Blurry	**0.0**	**0.0**	**0.0**	**0.0**	–	**0.0**	**0.0**	–	–	**0.0**	**0.0**	–	–	**0.0**
Brown_Hair	+0.7	–	**0.0**	+0.7	–	–	**0.0**	+0.7	-3.7	**0.0**	+0.7	–	–	**0.0**
Bushy_Eyebrows	+1.0	-0.7	+0.7	+3.7	**0.0**	–	**0.0**	+1.0	–	+0.7	+3.7	–	–	**+0.3**
Chubby	**0.0**	**0.0**	**0.0**	+0.3	–	+0.7	**0.0**	**0.0**	**0.0**	**0.0**	+0.3	**0.0**	–	**0.0**
Double_Chin	**0.0**	**0.0**	**0.0**	**0.0**	–	**0.0**	**0.0**	**0.0**	–	**0.0**	**0.0**	–	**0.0**	–
Eyeglasses	+2.7	**0.0**	**0.0**	+6.0	–	–	**0.0**	+2.7	–	**0.0**	+6.0	–	–	–
Goatee	**0.0**	**0.0**	**0.0**	-0.3	–	–	**0.0**	**0.0**	–	**0.0**	-0.3	–	–	–
Gray_Hair	**0.0**	**0.0**	**0.0**	+2.0	–	–	**0.0**	**0.0**	**0.0**	**0.0**	+2.0	–	–	**0.0**
Heavy_Makeup	+25.0	+26.0	+26.0	+23.7	**0.0**	-4.0	+0.7	–	–	**+26.3**	**+26.3**	–	–	–
High_Cheekbones	+21.7	+18.7	+21.3	+14.7	–	–	**-0.3**	–	–	**+25.0**	+27.0	–	–	–
Male	+37.3	+37.7	+36.3	+33.7	**0.0**	**0.0**	-0.3	+37.3	–	**+36.3**	+37.0	–	–	–
Mouth_Slightly_Open	+13.7	+11.7	+14.0	+5.3	–	–	**0.0**	–	–	+14.0	**+8.7**	–	–	–
Mustache	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	–	–	**0.0**
Narrow_Eyes	**0.0**	**0.0**	**0.0**	**0.0**	–	–	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	–	–	**0.0**
No_Beard	+7.0	+1.7	+5.7	+8.0	**0.0**	**0.0**	**0.0**	+7.0	+1.7	+5.7	+8.0	–	–	**+0.3**
Oval_Face	+0.3	–	**0.0**	–	–	–	–	–	–	**0.0**	**0.0**	–	–	**0.0**
Pale_Skin	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	–	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	–	**0.0**
Pointy_Nose	-5.0	-7.3	-0.3	+2.0	–	**0.0**	**0.0**	–	–	**-0.3**	+2.0	–	-2.0	–
Receding_Hairline	–	**0.0**	**0.0**	–	–	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	+0.3	–	–	**0.0**
Rosy_Cheeks	+0.3	**0.0**	**0.0**	+1.3	**0.0**	**0.0**	**0.0**	–	–	**0.0**	+1.3	–	–	–
Sideburns	+0.3	**0.0**	**0.0**	+0.7	–	**0.0**	**0.0**	+0.3	**0.0**	**0.0**	+0.7	–	–	**0.0**
Smiling	+30.3	+26.7	+22.3	+16.0	–	+25.3	**0.0**	–	–	**+29.3**	+34.3	–	–	–
Straight_Hair	**0.0**	+0.3	**0.0**	+2.3	–	**0.0**	**0.0**	**0.0**	+0.3	**0.0**	+2.3	–	–	**0.0**
Wavy_Hair	+14.0	+10.7	+8.0	+15.3	**0.0**	**0.0**	**0.0**	–	–	**+8.0**	+15.3	–	–	–
Wearing_Earrings	+2.7	**0.0**	+0.3	+6.7	**0.0**	**0.0**	**0.0**	–	–	**+0.3**	+6.7	–	–	–
Wearing_Hat	**0.0**	**0.0**	**0.0**	+2.3	–	**0.0**	**0.0**	–	–	**0.0**	+2.3	–	–	**0.0**
Wearing_Lipstick	+40.0	+38.0	+39.0	+27.7	–	–	**+0.3**	–	–	+39.0	**+38.3**	–	–	–
Wearing_Necklace	-1.0	**0.0**	**0.0**	**0.0**	–	–	**0.0**	–	–	**0.0**	**0.0**	–	–	–
Wearing_Necktie	+1.3	**0.0**	**0.0**	+2.0	**0.0**	–	**0.0**	+1.3	–	**0.0**	+2.0	–	–	–
Young	+11.0	+9.3	+11.0	+12.3	**0.0**	–	+2.7	+11.0	–	+11.0	+12.3	–	**+8.7**	–
Mean (pp)	+6.6	+5.4	+5.5	+5.8	0.0	+1.4	+0.2	+3.9	+0.5	+5.8	+7.0	0.0	+2.2	+0.4
TanGCE^{++} W/T/L				6/30/3							1/14/1			

Table 7: **CelebA per-attribute breakdown at $\Delta Y \leq 3$pp (40 attributes $\times$ 2 control regimes).** Signed $(S - S_{\text{floor}})$ in pp at the trajectory step closest to the per-attribute majority-class floor, subject to mean $\Delta Y \geq -3$pp on the 5 controls. Bold = best (smallest absolute deviation from floor) per attribute per regime block. '–' = no in-budget step at that budget for that method. Final rows: per-regime per-method mean and TanGCE^{++} W/T/L counts vs. the best other method (0.5pp tie band).

Concept	5 least-correlated controls							5 most-correlated controls						
	LE	L+C	INLP	IGBP	Obl	Amb++	Tan++	LE	L+C	INLP	IGBP	Obl	Amb++	Tan++
5_o_Clock_Shadow	+3.0	+1.3	**0.0**	+3.0	**0.0**	**0.0**	**0.0**	+3.0	+1.3	**0.0**	+3.0	–	–	**0.0**
Arched_Eyebrows	+9.7	+3.0	+6.7	+6.7	–	+4.7	**0.0**	–	+3.0	+6.7	+6.7	–	–	**0.0**
Attractive	+15.3	+17.3	+17.7	+1.0	–	–	**-0.3**	+15.3	–	+17.0	**+1.0**	–	–	–
Bags_Under_Eyes	+0.3	**0.0**	**0.0**	+0.3	–	**0.0**	**0.0**	+0.3	–	**0.0**	+0.3	**0.0**	–	–
Bald	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Bangs	+4.7	**0.0**	**0.0**	+3.7	**0.0**	+1.7	**0.0**	+4.7	**0.0**	**0.0**	+3.7	**0.0**	–	**0.0**
Big_Lips	+1.7	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	+1.7	**0.0**	**0.0**	**0.0**	–	–	**0.0**
Big_Nose	+2.3	+2.3	**0.0**	+2.0	–	+1.3	**0.0**	+2.3	+2.3	**0.0**	+2.0	–	+1.3	-0.3
Black_Hair	+10.7	+7.7	+8.3	+4.0	–	+6.7	**+1.3**	+10.7	+7.7	+8.3	+4.0	**0.0**	+6.7	+1.3
Blond_Hair	+5.0	+3.0	+3.0	+3.3	**0.0**	**0.0**	**0.0**	+5.0	+3.0	+3.0	+3.3	–	+5.3	**+0.3**
Blurry	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	–	**0.0**	**0.0**
Brown_Hair	+0.7	-3.7	**0.0**	+0.7	**0.0**	-0.3	**0.0**	+0.7	-3.7	**0.0**	+0.7	–	–	**0.0**
Bushy_Eyebrows	+1.0	-0.7	+0.7	+3.7	**0.0**	-0.3	**0.0**	+1.0	-0.7	+0.7	+3.7	–	–	**0.0**
Chubby	**0.0**	**0.0**	**0.0**	+0.3	**0.0**	+0.7	**0.0**	**0.0**	**0.0**	**0.0**	+0.3	**0.0**	**0.0**	**0.0**
Double_Chin	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	–	**0.0**	**0.0**
Eyeglasses	+2.7	**0.0**	**0.0**	+6.0	–	**0.0**	**0.0**	+2.7	**0.0**	**0.0**	+6.0	**0.0**	–	**0.0**
Goatee	**0.0**	**0.0**	**0.0**	-0.3	–	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	-0.3	–	–	**0.0**
Gray_Hair	**0.0**	**0.0**	**0.0**	+2.0	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	+2.0	**0.0**	**0.0**	**0.0**
Heavy_Makeup	+25.0	+26.0	+26.0	+23.7	**0.0**	**0.0**	**0.0**	+25.0	+26.0	+26.0	**+23.7**	–	+25.3	–
High_Cheekbones	+21.7	+18.7	+21.3	+14.7	**0.0**	+17.0	-0.3	–	–	+23.3	**+14.7**	–	–	–
Male	+37.3	+37.7	+36.3	+33.7	**0.0**	**0.0**	-0.3	+37.3	+37.7	+36.3	**+34.0**	–	+37.3	+36.0
Mouth_Slightly_Open	+13.7	+11.7	+14.0	+5.3	–	+12.7	**0.0**	+13.7	+11.7	+14.0	**+5.3**	–	–	+6.3
Mustache	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	–	**0.0**
Narrow_Eyes	**0.0**	**0.0**	**0.0**	**0.0**	–	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
No_Beard	+7.0	+1.7	+5.7	+8.0	**0.0**	**0.0**	**0.0**	+7.0	+1.7	+5.7	+8.0	**0.0**	+2.3	-0.3
Oval_Face	+0.3	**0.0**	**0.0**	**0.0**	**0.0**	-4.0	**0.0**	+0.3	**0.0**	**0.0**	**0.0**	**0.0**	–	**0.0**
Pale_Skin	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	–	**0.0**
Pointy_Nose	-5.0	-7.3	-0.3	+2.0	**0.0**	**0.0**	**0.0**	–	–	**-0.3**	+2.0	–	-2.0	–
Receding_Hairline	**0.0**	**0.0**	**0.0**	+0.3	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	+0.3	**0.0**	**0.0**	**0.0**
Rosy_Cheeks	+0.3	**0.0**	**0.0**	+1.3	**0.0**	**0.0**	**0.0**	+0.3	**0.0**	**0.0**	+1.3	–	–	**0.0**
Sideburns	+0.3	**0.0**	**0.0**	+0.7	**0.0**	**0.0**	**0.0**	+0.3	**0.0**	**0.0**	+0.7	**0.0**	-0.3	**0.0**
Smiling	+30.3	+26.7	+22.3	+16.0	–	+25.3	**0.0**	+30.3	–	**+24.3**	+25.0	–	+25.3	–
Straight_Hair	**0.0**	+0.3	**0.0**	+2.3	**0.0**	**0.0**	**0.0**	**0.0**	+0.3	**0.0**	+2.3	–	+0.3	**0.0**
Wavy_Hair	+14.0	+10.7	+8.0	+15.3	**0.0**	**0.0**	**0.0**	+14.0	–	+8.0	+15.3	–	–	**+3.7**
Wearing_Earrings	+2.7	**0.0**	+0.3	+6.7	**0.0**	**0.0**	**0.0**	+2.7	**0.0**	+0.3	+6.7	–	–	**0.0**
Wearing_Hat	**0.0**	**0.0**	**0.0**	+2.3	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	+2.3	**0.0**	**0.0**	**0.0**
Wearing_Lipstick	+40.0	+38.0	+39.0	+27.7	–	+38.3	**+0.3**	+40.0	+38.0	+39.0	**+31.3**	–	+38.3	+33.0
Wearing_Necklace	-1.0	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	-1.0	**0.0**	**0.0**	**0.0**	–	–	**0.0**
Wearing_Necktie	+1.3	**0.0**	**0.0**	+2.0	**0.0**	**0.0**	**0.0**	+1.3	**0.0**	**0.0**	+2.0	–	–	**0.0**
Young	+11.0	+9.3	+11.0	+12.3	**0.0**	+8.7	**0.0**	+11.0	+9.3	+11.0	+12.3	–	+8.7	**+0.3**
Mean (pp)	+6.4	+5.1	+5.5	+5.3	0.0	+2.9	0.0	+6.2	+4.0	+5.6	+5.6	0.0	+7.4	+2.4
TanGCE^{++} W/T/L				6/34/0							5/25/4			

Table 8: **CelebA per-attribute breakdown at $\Delta Y \leq 5$pp (40 attributes $\times$ 2 control regimes).** Signed $(S - S_{\text{floor}})$ in pp at the trajectory step closest to the per-attribute majority-class floor, subject to mean $\Delta Y \geq -5$pp on the 5 controls. Bold = best (smallest absolute deviation from floor) per attribute per regime block. '–' = no in-budget step at that budget for that method. Final rows: per-regime per-method mean and TanGCE^{++} W/T/L counts vs. the best other method (0.5pp tie band).

Concept	5 least-correlated controls							5 most-correlated controls						
	LE	L+C	INLP	IGBP	Obl	Amb++	Tan++	LE	L+C	INLP	IGBP	Obl	Amb++	Tan++
5_o_Clock_Shadow	+3.0	+1.3	**0.0**	+3.0	**0.0**	**0.0**	**0.0**	+3.0	+1.3	**0.0**	+3.0	–	+1.3	**0.0**
Arched_Eyebrows	+9.7	+3.0	+6.7	+6.7	**0.0**	-1.7	**0.0**	+9.7	+3.0	+6.7	+6.7	–	+4.7	**0.0**
Attractive	+15.3	+17.3	+17.7	+1.0	-12.0	+15.7	**-0.3**	+15.3	+17.3	+17.0	**+1.0**	–	+16.0	+13.7
Bags_Under_Eyes	+0.3	**0.0**	**0.0**	+0.3	**0.0**	**0.0**	**0.0**	+0.3	**0.0**	**0.0**	+0.3	**0.0**	**0.0**	**0.0**
Bald	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Bangs	+4.7	**0.0**	**0.0**	+3.7	**0.0**	+1.7	**0.0**	+4.7	**0.0**	**0.0**	+3.7	**0.0**	+1.7	**0.0**
Big_Lips	+1.7	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	+1.7	**0.0**	**0.0**	**0.0**	**0.0**	+0.7	**0.0**
Big_Nose	+2.3	+2.3	**0.0**	+2.0	**0.0**	**0.0**	**0.0**	+2.3	+2.3	**0.0**	+2.0	**0.0**	**0.0**	-0.3
Black_Hair	+10.7	+7.7	+8.3	+4.0	**0.0**	+6.7	+0.7	+10.7	+7.7	+8.3	+4.0	**0.0**	**0.0**	+1.0
Blond_Hair	+5.0	+3.0	+3.0	+3.3	**0.0**	**0.0**	**0.0**	+5.0	+3.0	+3.0	+3.3	**0.0**	+5.3	**0.0**
Blurry	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Brown_Hair	+0.7	-3.7	**0.0**	+0.7	**0.0**	-0.3	**0.0**	+0.7	-3.7	**0.0**	+0.7	**0.0**	–	**0.0**
Bushy_Eyebrows	+1.0	-0.7	+0.7	+3.7	**0.0**	-0.3	**0.0**	+1.0	-0.7	+0.7	+3.7	–	–	**0.0**
Chubby	**0.0**	**0.0**	**0.0**	+0.3	**0.0**	+0.7	**0.0**	**0.0**	**0.0**	**0.0**	+0.3	**0.0**	**0.0**	**0.0**
Double_Chin	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Eyeglasses	+2.7	**0.0**	**0.0**	+6.0	**0.0**	**0.0**	**0.0**	+2.7	**0.0**	**0.0**	+6.0	**0.0**	–	**0.0**
Goatee	**0.0**	**0.0**	**0.0**	-0.3	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	-0.3	**0.0**	–	**0.0**
Gray_Hair	**0.0**	**0.0**	**0.0**	+2.0	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	+2.0	**0.0**	**0.0**	**0.0**
Heavy_Makeup	+25.0	+26.0	+26.0	+23.7	**0.0**	**0.0**	**0.0**	+25.0	+26.0	+26.0	+23.7	–	+25.3	**+17.7**
High_Cheekbones	+21.7	+18.7	+21.3	+14.7	**0.0**	**0.0**	-0.3	+21.7	–	+21.3	**+14.7**	–	+17.0	–
Male	+37.3	+37.7	+36.3	+33.7	**0.0**	**0.0**	-0.3	+37.3	+37.7	+36.3	**+33.0**	–	+37.3	+33.7
Mouth_Slightly_Open	+13.7	+11.7	+14.0	+5.3	**0.0**	+0.7	**0.0**	+13.7	+11.7	+14.0	+5.3	**0.0**	+12.7	**0.0**
Mustache	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	–	**0.0**
Narrow_Eyes	**0.0**	**0.0**	**0.0**	**0.0**	–	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
No_Beard	+7.0	+1.7	+5.7	+8.0	**0.0**	**0.0**	**0.0**	+7.0	+1.7	+5.7	+8.0	**0.0**	+2.3	**0.0**
Oval_Face	+0.3	**0.0**	**0.0**	**0.0**	**0.0**	-1.7	**0.0**	+0.3	**0.0**	**0.0**	**0.0**	**0.0**	-4.0	**0.0**
Pale_Skin	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Pointy_Nose	-5.0	-7.3	-0.3	+2.0	**0.0**	**0.0**	**0.0**	-5.0	-7.3	-0.3	+2.0	**0.0**	-2.0	**0.0**
Receding_Hairline	**0.0**	**0.0**	**0.0**	+0.3	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	+0.3	**0.0**	**0.0**	**0.0**
Rosy_Cheeks	+0.3	**0.0**	**0.0**	+1.3	**0.0**	**0.0**	**0.0**	+0.3	**0.0**	**0.0**	+1.3	**0.0**	–	**0.0**
Sideburns	+0.3	**0.0**	**0.0**	+0.7	**0.0**	**0.0**	**0.0**	+0.3	**0.0**	**0.0**	+0.7	**0.0**	-0.3	**0.0**
Smiling	+30.3	+26.7	+22.3	+16.0	**0.0**	-0.3	**0.0**	+30.3	+26.7	+22.3	**+17.3**	–	+25.3	–
Straight_Hair	**0.0**	+0.3	**0.0**	+2.3	**0.0**	**0.0**	**0.0**	**0.0**	+0.3	**0.0**	+2.3	–	+0.3	**0.0**
Wavy_Hair	+14.0	+10.7	+8.0	+15.3	**0.0**	**0.0**	**0.0**	+14.0	+10.7	+8.0	+15.3	**0.0**	–	-0.3
Wearing_Earrings	+2.7	**0.0**	+0.3	+6.7	**0.0**	**0.0**	**0.0**	+2.7	**0.0**	+0.3	+6.7	**0.0**	**0.0**	**0.0**
Wearing_Hat	**0.0**	**0.0**	**0.0**	+2.3	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	+2.3	**0.0**	**0.0**	**0.0**
Wearing_Lipstick	+40.0	+38.0	+39.0	+27.7	-4.3	**0.0**	+0.3	+40.0	+38.0	+39.0	**+30.0**	–	+38.3	+33.0
Wearing_Necklace	-1.0	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	-1.0	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Wearing_Necktie	+1.3	**0.0**	**0.0**	+2.0	**0.0**	**0.0**	**0.0**	+1.3	**0.0**	**0.0**	+2.0	**0.0**	**0.0**	**0.0**
Young	+11.0	+9.3	+11.0	+12.3	**0.0**	+8.7	**0.0**	+11.0	+9.3	+11.0	+12.3	**0.0**	+8.7	+0.3
Mean (pp)	+6.4	+5.1	+5.5	+5.3	-0.4	+0.7	0.0	+6.4	+4.7	+5.5	+5.3	0.0	+5.8	+2.6
TanGCE^{++} W/T/L				1/38/1							3/31/4			

Table 9: **CelebA least-correlated control sets (low-entanglement regime).** For each of the 40 target attributes (rows), the 5 control attributes selected as the bottom-5 by absolute Pearson $|r|$ on the train split, with each $|r|$ value in parentheses. These are the 5 attributes that share the least information with the target and isolate clean surgicality.

Target	5 least-correlated controls (with $\|r\|$)				
5_o_Clock_Shadow	Receding_Hairline (0.00)	Double_Chin (0.01)	Young (0.01)	Bald (0.01)	Blurry (0.02)
Arched_Eyebrows	Oval_Face (0.00)	Bushy_Eyebrows (0.01)	Bangs (0.01)	Receding_Hairline (0.02)	Black_Hair (0.03)
Attractive	Straight_Hair (0.00)	Bushy_Eyebrows (0.01)	Black_Hair (0.01)	Mouth_Slightly_Open (0.04)	Wearing_Necklace (0.08)
Bags_Under_Eyes	Straight_Hair (0.00)	Wearing_Necklace (0.01)	Big_Lips (0.01)	Wearing_Hat (0.01)	Black_Hair (0.02)
Bald	Wearing_Hat (0.00)	Mouth_Slightly_Open (0.01)	Smiling (0.01)	5_o_Clock_Shadow (0.01)	High_Cheekbones (0.03)
Bangs	Black_Hair (0.00)	Brown_Hair (0.01)	Mouth_Slightly_Open (0.01)	Narrow_Eyes (0.01)	Arched_Eyebrows (0.01)
Big_Lips	Chubby (0.00)	Mustache (0.00)	Goatee (0.00)	Bags_Under_Eyes (0.01)	Double_Chin (0.01)
Big_Nose	Wearing_Necklace (0.01)	Straight_Hair (0.03)	Rosy_Cheeks (0.03)	Wearing_Earrings (0.03)	Black_Hair (0.05)
Black_Hair	Bangs (0.00)	Smiling (0.00)	Mouth_Slightly_Open (0.01)	Attractive (0.01)	Chubby (0.01)
Blond_Hair	Blurry (0.00)	Straight_Hair (0.01)	Pale_Skin (0.03)	Oval_Face (0.04)	Narrow_Eyes (0.05)
Blurry	Blond_Hair (0.00)	Sideburns (0.00)	Goatee (0.00)	No_Beard (0.01)	Wearing_Hat (0.01)
Brown_Hair	Wearing_Necklace (0.00)	Mouth_Slightly_Open (0.00)	Bangs (0.01)	Straight_Hair (0.02)	High_Cheekbones (0.02)
Bushy_Eyebrows	Chubby (0.00)	Narrow_Eyes (0.00)	Smiling (0.00)	Wearing_Hat (0.01)	Arched_Eyebrows (0.01)
Chubby	Big_Lips (0.00)	Bushy_Eyebrows (0.00)	Mouth_Slightly_Open (0.01)	Wearing_Necklace (0.01)	High_Cheekbones (0.01)
Double_Chin	5_o_Clock_Shadow (0.01)	Rosy_Cheeks (0.01)	Wearing_Necklace (0.01)	Big_Lips (0.01)	Sideburns (0.02)
Eyeglasses	Straight_Hair (0.01)	Blurry (0.01)	Black_Hair (0.02)	Narrow_Eyes (0.03)	Pale_Skin (0.03)
Goatee	Blurry (0.00)	Big_Lips (0.00)	Double_Chin (0.02)	Receding_Hairline (0.03)	Black_Hair (0.03)
Gray_Hair	Sideburns (0.01)	Narrow_Eyes (0.02)	Bushy_Eyebrows (0.02)	Smiling (0.02)	Straight_Hair (0.03)
Heavy_Makeup	Pale_Skin (0.07)	Receding_Hairline (0.07)	Straight_Hair (0.07)	Narrow_Eyes (0.07)	Brown_Hair (0.10)
High_Cheekbones	Chubby (0.01)	Straight_Hair (0.02)	Brown_Hair (0.02)	Young (0.02)	Bushy_Eyebrows (0.02)
Male	Narrow_Eyes (0.02)	Blurry (0.02)	Straight_Hair (0.05)	Receding_Hairline (0.08)	Pale_Skin (0.10)
Mouth_Slightly_Open	Brown_Hair (0.00)	Chubby (0.01)	Bangs (0.01)	Bald (0.01)	Black_Hair (0.01)
Mustache	Big_Lips (0.00)	Pale_Skin (0.01)	Blurry (0.01)	Narrow_Eyes (0.02)	Bald (0.03)
Narrow_Eyes	Wearing_Earrings (0.00)	Bushy_Eyebrows (0.00)	Sideburns (0.00)	Bangs (0.01)	No_Beard (0.01)
No_Beard	Blurry (0.01)	Narrow_Eyes (0.01)	Straight_Hair (0.02)	Double_Chin (0.03)	Receding_Hairline (0.04)
Oval_Face	Pointy_Nose (0.00)	Straight_Hair (0.00)	Arched_Eyebrows (0.00)	Receding_Hairline (0.01)	Wavy_Hair (0.01)
Pale_Skin	Wearing_Hat (0.00)	Mustache (0.01)	Wearing_Necktie (0.02)	Wavy_Hair (0.02)	Bushy_Eyebrows (0.02)
Pointy_Nose	Oval_Face (0.00)	Straight_Hair (0.00)	Receding_Hairline (0.01)	5_o_Clock_Shadow (0.03)	Pale_Skin (0.03)
Receding_Hairline	5_o_Clock_Shadow (0.00)	Rosy_Cheeks (0.00)	Straight_Hair (0.01)	Oval_Face (0.01)	Pointy_Nose (0.01)
Rosy_Cheeks	Receding_Hairline (0.00)	Double_Chin (0.01)	Chubby (0.02)	Narrow_Eyes (0.03)	Bushy_Eyebrows (0.03)
Sideburns	Blurry (0.00)	Narrow_Eyes (0.00)	Gray_Hair (0.01)	Straight_Hair (0.01)	Wearing_Necktie (0.01)
Smiling	Bushy_Eyebrows (0.00)	Black_Hair (0.00)	Straight_Hair (0.00)	Bald (0.01)	Brown_Hair (0.02)
Straight_Hair	Oval_Face (0.00)	Bags_Under_Eyes (0.00)	Wearing_Necklace (0.00)	Pointy_Nose (0.00)	Smiling (0.00)
Wavy_Hair	Oval_Face (0.01)	Pale_Skin (0.02)	Narrow_Eyes (0.02)	Double_Chin (0.02)	Sideburns (0.03)
Wearing_Earrings	Narrow_Eyes (0.00)	Receding_Hairline (0.02)	Double_Chin (0.02)	Chubby (0.03)	Oval_Face (0.03)
Wearing_Hat	Pale_Skin (0.00)	Bald (0.00)	Bushy_Eyebrows (0.01)	Wearing_Necktie (0.01)	Bags_Under_Eyes (0.01)
Wearing_Lipstick	Straight_Hair (0.04)	Narrow_Eyes (0.05)	Pale_Skin (0.08)	Receding_Hairline (0.09)	Black_Hair (0.11)
Wearing_Necklace	Brown_Hair (0.00)	Straight_Hair (0.00)	Bags_Under_Eyes (0.01)	Big_Nose (0.01)	Double_Chin (0.01)
Wearing_Necktie	Wearing_Hat (0.01)	Sideburns (0.01)	Pale_Skin (0.02)	Blurry (0.02)	Smiling (0.03)
Young	5_o_Clock_Shadow (0.01)	Wearing_Necklace (0.02)	Mouth_Slightly_Open (0.02)	High_Cheekbones (0.02)	Wearing_Hat (0.03)

Table 10: **CelebA most-correlated control sets (high-entanglement regime).** For each of the 40 target attributes (rows), the 5 control attributes selected as the top-5 by absolute Pearson $|r|$ on the train split, with each $|r|$ value in parentheses. These share substantial information with the target and form the surgicality stress test.

Target	5 most-correlated controls (with $\lvert r \rvert$)				
5_o_Clock_Shadow	No_Beard (0.53)	Male (0.42)	Wearing_Lipstick (0.35)	Heavy_Makeup (0.29)	Sideburns (0.27)
Arched_Eyebrows	Wearing_Lipstick (0.46)	Heavy_Makeup (0.44)	Male (0.42)	Attractive (0.28)	Big_Lips (0.28)
Attractive	Wearing_Lipstick (0.51)	Heavy_Makeup (0.50)	Male (0.45)	Young (0.42)	Arched_Eyebrows (0.28)
Bags_Under_Eyes	Big_Nose (0.35)	Male (0.30)	Heavy_Makeup (0.28)	Wearing_Lipstick (0.26)	Young (0.25)
Bald	Gray_Hair (0.28)	Young (0.22)	Chubby (0.22)	Big_Nose (0.22)	Wearing_Necktie (0.19)
Bangs	Male (0.20)	Wearing_Lipstick (0.20)	No_Beard (0.15)	Heavy_Makeup (0.14)	Receding_Hairline (0.13)
Big_Lips	Arched_Eyebrows (0.28)	Male (0.21)	Wearing_Lipstick (0.21)	Wavy_Hair (0.18)	Wearing_Necklace (0.17)
Big_Nose	Male (0.36)	Bags_Under_Eyes (0.35)	Young (0.33)	Double_Chin (0.31)	Chubby (0.29)
Black_Hair	Bushy_Eyebrows (0.29)	Brown_Hair (0.24)	Blond_Hair (0.24)	Young (0.15)	5_o_Clock_Shadow (0.14)
Blond_Hair	Male (0.31)	Wearing_Lipstick (0.31)	Heavy_Makeup (0.28)	Black_Hair (0.24)	Arched_Eyebrows (0.20)
Blurry	Attractive (0.18)	Heavy_Makeup (0.16)	Wearing_Lipstick (0.15)	Young (0.11)	Arched_Eyebrows (0.09)
Brown_Hair	Black_Hair (0.24)	Blond_Hair (0.17)	Male (0.14)	Wearing_Lipstick (0.13)	Wavy_Hair (0.13)
Bushy_Eyebrows	Black_Hair (0.29)	No_Beard (0.25)	Male (0.23)	5_o_Clock_Shadow (0.17)	Wearing_Lipstick (0.16)
Chubby	Double_Chin (0.46)	Young (0.30)	Big_Nose (0.29)	Wearing_Necktie (0.22)	Bald (0.22)
Double_Chin	Chubby (0.46)	Young (0.33)	Big_Nose (0.31)	Gray_Hair (0.25)	Bags_Under_Eyes (0.24)
Eyeglasses	Wearing_Lipstick (0.24)	Chubby (0.21)	Male (0.21)	Heavy_Makeup (0.20)	Attractive (0.20)
Goatee	No_Beard (0.56)	Sideburns (0.51)	Mustache (0.43)	Male (0.30)	Wearing_Lipstick (0.26)
Gray_Hair	Young (0.35)	Bald (0.28)	Double_Chin (0.25)	Wearing_Necktie (0.23)	Receding_Hairline (0.23)
Heavy_Makeup	Wearing_Lipstick (0.82)	Male (0.68)	Attractive (0.50)	Arched_Eyebrows (0.44)	Wearing_Earrings (0.37)
High_Cheekbones	Smiling (0.69)	Mouth_Slightly_Open (0.44)	Wearing_Lipstick (0.29)	Wearing_Earrings (0.29)	Male (0.27)
Male	Wearing_Lipstick (0.80)	Heavy_Makeup (0.68)	No_Beard (0.52)	Attractive (0.45)	5_o_Clock_Shadow (0.42)
Mouth_Slightly_Open	Smiling (0.53)	High_Cheekbones (0.44)	Wearing_Earrings (0.22)	Rosy_Cheeks (0.15)	Arched_Eyebrows (0.13)
Mustache	Goatee (0.43)	No_Beard (0.43)	Sideburns (0.36)	Male (0.24)	Big_Nose (0.20)
Narrow_Eyes	Big_Lips (0.16)	Oval_Face (0.14)	Big_Nose (0.13)	Double_Chin (0.12)	Mouth_Slightly_Open (0.09)
No_Beard	Goatee (0.56)	5_o_Clock_Shadow (0.53)	Sideburns (0.52)	Male (0.52)	Wearing_Lipstick (0.43)
Oval_Face	Heavy_Makeup (0.22)	High_Cheekbones (0.21)	Attractive (0.21)	Smiling (0.20)	Rosy_Cheeks (0.17)
Pale_Skin	Attractive (0.12)	Male (0.10)	Young (0.09)	High_Cheekbones (0.09)	Smiling (0.09)
Pointy_Nose	Wearing_Lipstick (0.25)	Heavy_Makeup (0.25)	Male (0.24)	Arched_Eyebrows (0.23)	Attractive (0.18)
Receding_Hairline	Gray_Hair (0.23)	Big_Nose (0.22)	Young (0.18)	Double_Chin (0.15)	Bags_Under_Eyes (0.13)
Rosy_Cheeks	Heavy_Makeup (0.28)	Arched_Eyebrows (0.26)	Wearing_Lipstick (0.25)	High_Cheekbones (0.25)	Smiling (0.24)
Sideburns	No_Beard (0.52)	Goatee (0.51)	Mustache (0.36)	Male (0.28)	5_o_Clock_Shadow (0.27)
Smiling	High_Cheekbones (0.69)	Mouth_Slightly_Open (0.53)	Rosy_Cheeks (0.24)	Wearing_Earrings (0.23)	Wearing_Lipstick (0.22)
Straight_Hair	Wavy_Hair (0.31)	Black_Hair (0.14)	Wearing_Hat (0.10)	Wearing_Earrings (0.09)	Blurry (0.09)
Wavy_Hair	Wearing_Lipstick (0.35)	Heavy_Makeup (0.33)	Male (0.33)	Straight_Hair (0.31)	Attractive (0.24)
Wearing_Earrings	Wearing_Lipstick (0.37)	Heavy_Makeup (0.37)	Male (0.37)	High_Cheekbones (0.29)	Arched_Eyebrows (0.26)
Wearing_Hat	Wearing_Lipstick (0.17)	Heavy_Makeup (0.17)	Attractive (0.13)	Big_Nose (0.12)	Male (0.12)
Wearing_Lipstick	Heavy_Makeup (0.82)	Male (0.80)	Attractive (0.51)	Arched_Eyebrows (0.46)	No_Beard (0.43)
Wearing_Necklace	Male (0.26)	Wearing_Lipstick (0.24)	Arched_Eyebrows (0.23)	Wearing_Earrings (0.17)	Big_Lips (0.17)
Wearing_Necktie	Male (0.30)	Wearing_Lipstick (0.24)	Young (0.24)	Gray_Hair (0.23)	Big_Nose (0.22)
Young	Attractive (0.42)	Gray_Hair (0.35)	Double_Chin (0.33)	Big_Nose (0.33)	Male (0.32)

Table 11: **Per-row λ_i statistics for the proposed method ($\varepsilon = 0.1$, $\lambda_{\max} = 64$) on the 39-setting NLP grid at 50%-depth.** Each cell averages the per-row λ_i first across all 60 outer iterations of one trajectory, then across the 13 model settings of the concept. The within-setting p_{10}/p_{90} spread (mean across settings) is roughly $14/45$, i.e. a $\sim$10$\times$ ratio between the least- and most-confident rows in a single iteration. The safety cap $\lambda_{\max} = 64$ is observed only sporadically (max 57.32 across 39$\times$60 = 2340 iterations); the closed-form rule rarely binds against it.

Concept	n	mean λ_i	median	min	max
Sycophancy	13	29.58	37.15	4.89	49.38
Gender	13	25.28	23.76	0.45	45.08
Safety	13	33.07	33.30	1.71	57.32
Overall	**39**	**29.31**	**30.19**	**0.45**	**57.32**